ESSENTIAL MATHEMATICS GRADE 10: REAL NUMBERS

CBE Phenomenon-Driven Lesson Sequence — Sub-Strand 1.1 (8 Lessons)

SUB-STRAND OVERVIEW	
Grade Level	10
Subject	Essential Mathematics
Strand	Strand 1.0: Numbers and Algebra
Sub-Strand	Sub-Strand 1.1: Real Numbers
Total Duration	8 lessons × 40 minutes = 320 minutes total
Sub-Strand Content	– Classification of real numbers as rational or irrational – Odd, even, prime and composite numbers – Combined operations on rational numbers (order of operations) – Reciprocal of real numbers (using calculators) – Application of integers and rational numbers in real-life situations (e.g., body temperature, blood pressure, shoe sizes)
Learning Outcomes	By the end of the sub-strand, the learner should be able to: a) classify real numbers as rational or irrational, b) perform combined operations on rational numbers, c) determine the reciprocal of real numbers using calculators, d) apply rational numbers in real life situations, e) appreciate the use of real numbers in real life.
Core Competencies	– Critical Thinking and Problem Solving: the learner develops interpretation and inference skills as they classify numbers as either rational or irrational numbers. – Self-efficacy: the learner develops self-awareness as they discuss the use of real numbers in real life such as in body temperature and blood pressure.
Core Values	– Responsibility: the learner takes care of a calculator when using it to work out reciprocal of numbers. – Respect: the learner develops humility by taking turns while discussing with peers and working out combined operations.
Science & Engineering Practices	– Asking Questions and Defining Problems: learners pose and refine questions about why certain measurements (e.g., a Nairobi market vendor's daily tally) produce rational vs. irrational results, driving the inquiry into real-number classification. – Analysing and Interpreting Data: learners collect personal biometric data (weight, height, shoe size) and analyse whether each measurement is best represented by a rational or irrational number. – Using Mathematics and Computational Thinking: learners apply order-of-operations rules and calculator-based reciprocal computation to solve multi-step problems set in Kenyan market and health contexts. – Constructing Explanations: learners build written and verbal explanations for why operations on rational numbers always yield rational results, anchored in the observed market-transaction phenomenon.

Pertinent & Contemporary Issues (PCIs)	– Health Promotion: the learner develops healthy practice skills while discussing the use of real numbers in recording body temperature and blood pressure — contextualised through scenarios at a Kenyan community health centre.
Career Connections	– Healthcare / Nursing: recording and interpreting patient vitals (temperature 36.8 °C, blood pressure 120/80) requires fluency with rational numbers and combined operations. – Market Trading / Sole Proprietorship: Mama mboga vendors at Gikomba Market use rational numbers daily for pricing, giving change, and calculating profit margins. – Sports Coaching / Athletics: timing and ranking Kenyan runners (e.g., at Kipchoge Keino Stadium, Eldoret) involves precise rational-number comparisons and averages. – Journalism / Media: reporters presenting poll results or economic statistics need to classify and manipulate rational numbers accurately for public communication.
Focus for Lessons	This unit grounds learners in the complete landscape of real numbers — rational and irrational — and builds confident fluency with combined operations and reciprocals in everyday Kenyan contexts. Using familiar anchors such as market transactions at Wakulima Market in Nairobi and health-clinic records from a rural dispensary in Kisumu County, learners move from intuitive number sense to precise classification and computation. The unit deliberately integrates calculator skills, peer discussion, and health-promotion awareness so that mathematical reasoning is always connected to real decisions learners and their communities make.
Driving Question / Key Inquiry Questions	DRIVING QUESTION: Why do some numbers from our everyday Kenyan lives "end neatly" while others go on forever — and how does knowing the difference help us make better decisions? KICD KEY INQUIRY QUESTIONS: 1. Why are numbers important?
Anchoring Phenomenon	A community health volunteer at the Mathare North Health Centre in Nairobi records the following patient data on a single morning: body temperatures of 36.8 °C, 37.0 °C, and 37.5 °C; blood pressure readings of 120/80 and 135/90; and shoe sizes of 40, 41, and 43. When she enters the data into a basic calculator to compute an average temperature, she notices that 36.8 + 37.0 + 37.5 divided by 3 gives a decimal that terminates exactly — yet when she tries to calculate the diagonal of the square floor tile (side = 1 m) in the waiting room using the Pythagorean theorem, the calculator displays a never-ending, non-repeating decimal ($\sqrt{2} \approx 1.41421356\ldots$). Students are puzzled: why do some of the health worker's numbers "end neatly" while others seem to go on forever? This tension — between numbers that can be written as exact fractions and those that cannot — anchors the entire sub-strand investigation into the nature of real numbers.
Storyline Thread	Lesson 1 — Anchoring Phenomenon & Number Classification (Predict Phase): Teacher presents the Mathare North Health Centre scenario. Learners individually record their own weight, height, and shoe size; classify each measurement as odd/even, prime/composite, and attempt to decide whether each is rational or irrational. Class creates a shared "What We Notice / What We Wonder" Driving Question Board (DQB). Cold-call 4 learners to share predictions about why the calculator diagonal "goes on forever." Lesson 2 — Observe: Rational vs. Irrational Numbers (Observe Phase): Learners examine a curated data set of Kenyan real-life numbers (maize prices, athletics times, tile diagonals, blood pressure values). In pairs they attempt to write each number as a fraction p/q; numbers that resist this are labelled irrational. Teacher guides a class sort on the board. Learners update their DQB with new evidence. Wait time: 60 seconds silent think before each cold-call (minimum 5 cold-calls per lesson). Lesson 3 — Explain: Defining and Formalising Rational vs. Irrational (Explain Phase): Learners formalise the definition of rational

	(p/q, q ≠ 0, p and q integers) and irrational numbers using worked examples from the health-centre phenomenon. Teacher models number-line placement of √2 and 22/7 (distinguishing the approximation from π). Learners complete a classification activity using 20 Kenyan-context numbers. DQB is updated with the formal definition. Lesson 4 — Combined Operations on Rational Numbers I (Observe + Explain): Using the sukuma wiki vendor scenario (supporting phenomenon 3), learners explore addition and subtraction of rational numbers including fractions and decimals. Teacher demonstrates BODMAS/PEMDAS with market-price contexts (ugali + beans + sukuma wiki total cost). Learners work in groups of three on case scenarios involving distribution of school stationery at a Kisumu school; each group presents one solution. DQB updated: "Do combined operations on rationals always give a rational result?" Lesson 5 — Combined Operations on Rational Numbers II — Multiplication & Division (Observe + Explain): Learners extend to multiplication and division of rational numbers through scenarios: a Kericho tea farmer divides 3¾ kg of dried tea leaves equally among 5 co-operative members; a Rift Valley maize farmer multiplies a seed rate of 2⅖ kg/acre over 3½ acres. Teacher facilitates a "gallery walk" — four worked problems posted around the room; pairs rotate (5 mins each) and annotate with corrections or alternative methods. Cold-call 3 pairs to defend their annotations. DQB updated. Lesson 6 — Reciprocals of Real Numbers Using Calculators (Observe + Explain): Teacher returns to the blood pressure reciprocal supporting phenomenon. Learners practise finding reciprocals of integers, fractions, and decimals using calculators; they record results in a table and identify the pattern (number × reciprocal = 1). Learners discuss: what is the reciprocal of an irrational number? Is it also irrational? Responsible calculator use (Responsibility value) is explicitly discussed. Learners update DQB. Lesson 7 — Real Numbers in Real-Life Applications & DQB Consolidation (Driving Question Board Creation): Learners work in groups on three application tasks: (a) computing a patient's average body temperature at a community dispensary; (b) calculating total and unit cost of githeri ingredients for a school canteen serving 200 learners; (c) determining whether a √5 m fence post length is rational or irrational and finding its decimal approximation to 4 s.f. Groups present findings; class revisits and expands the DQB, answering as many posted questions as possible with evidence gathered across lessons 1–6. Teacher cold-calls 6 learners to explain connections to the anchoring phenomenon. Lesson 8 — Model Building, Synthesis & Assessment (Model Building Phase): Each learner individually constructs a "Real Numbers Reference Map" — a hand-drawn or calculator-supported model showing the real-number system (naturals ⊂ integers ⊂ rationals ⊂ reals, with irrationals alongside), populated entirely with Kenyan-context examples from the unit. Learners annotate at least one combined-operations worked example and one reciprocal example on their map. Class discusses: "How has our answer to the driving question changed since Lesson 1?" Teacher administers a short class written test (portfolio evidence). Learners submit models as portfolio artefacts.
Supporting Phenomena	– Maize-sack weighing at Eldoret Grain Market: a grain trader weighs sacks and records 90 kg, 87.5 kg, and 91⅓ kg — all expressible as fractions — but the length of the rope needed to reach diagonally across a square 2 m × 2 m storage bay is √8 m, a number that cannot be expressed as a fraction. Why does the rope length "misbehave" while the weights do not? – Kenyan athletics timing boards at Kasarani Stadium: race times displayed as 9.58 s (rational) vs. the mathematical constant π used to compute the exact circumference of the circular track (irrational) — same track, two very different kinds of numbers. – Sukuma wiki price calculation: a vendor sells bunches at Ksh 5, Ksh 7.50, and Ksh 12.50 — combined operations on these rational prices always land on a rational total — inviting learners to test whether that pattern always holds. – Blood pressure reciprocal puzzle: a nurse is trained to consider the reciprocal of a patient's diastolic pressure ratio (1 ÷ 80 = 0.0125) when computing a medication dosage index — demonstrating a real-life need for reciprocals of rational numbers.

LESSON 1 (40 minutes): Anchoring Phenomenon & Number Classification — What Kind of Number Is That?	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To anchor the sub-strand investigation by presenting the Mathare North Health Centre phenomenon and activating learners' prior knowledge of number types, so that they can begin to sense the tension between numbers that terminate and numbers that go on forever.
Knowledge	Learners will know that numbers can be classified as odd, even, prime, composite, and — provisionally — as 'exact' or 'never-ending', forming an initial mental scaffold for the rational/irrational distinction explored across the sub-strand.
Skills	Learners will be able to record personal measurements (weight, height, shoe size), attempt a first classification of those measurements, and articulate a prediction about why √2 appears to be non-terminating on a calculator.
Attitudes	Learners will appreciate that numbers encountered in everyday Kenyan life — from a health worker's clinic register to the tiles on a waiting-room floor — are mathematically interesting and worth investigating deeply.
Key Inquiry Question	Why are numbers important?
Purpose in Storyline	This is the opening lesson of Sub-Strand 1.1. It launches the anchoring phenomenon (the Mathare North Health Centre scenario) and opens the Driving Question Board. Every subsequent lesson will return to this scenario to deepen understanding of the rational/irrational distinction.
Safety Notes	When learners record personal measurements such as weight and height, remind them to be sensitive to peers' emotions. KICD curriculum notes: 'learners to observe emotional issues when estimating weight and height.' Discourage comparison or ridicule; frame measurements as scientific data only.

B. LESSON OVERVIEW
Learners encounter the anchoring phenomenon: a community health volunteer at Mathare North Health Centre, Nairobi, records patient temperatures (36.8 °C, 37.0 °C, 37.5 °C), blood pressure readings, and shoe sizes, then notices that the average temperature divides neatly while the diagonal of the waiting-room floor tile (√2) produces a never-ending, non-repeating decimal. Working individually and then as a class, learners record their own weight, height, and shoe size; classify each as odd/even and prime/composite; and make their first attempt at labelling numbers as 'exact' or 'goes on forever'. The lesson closes with the creation of the class Driving Question Board (DQB) and cold-calling four learners to share predictions about the infinite decimal.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** [Prime numbers] Source: Khan Academy (English - US curriculum)	Learners listen as the teacher narrates the Mathare North Health Centre scenario (health volunteer, patient temperatures, blood pressure, shoe sizes, and the √2 tile diagonal). Each learner then	1. Display or read aloud the scenario card: 'Sister Wanjiku, a community health volunteer at Mathare North Health Centre, recorded these numbers this morning: temperatures 36.8 °C,	Prediction Journaling with Personal Data Anchoring — by using their own measurements as the first dataset, learners immediately connect abstract number types to personally	Circulate and look for two observable indicators: (a) Can the learner correctly classify at least one of their three measurements as odd/even? (b) Does the learner's prediction use any

🔍 Search ARES for similar videos 📖 **HTML:** Order of Real Numbers (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	silently records their own weight (kg), height (cm), and shoe size in their exercise book and writes a one-sentence prediction answering: 'Why does the calculator show a decimal that never ends for the diagonal, but the average temperature ends neatly?' Learners also attempt to classify their own measurements as odd/even and prime/composite, noting any difficulties.	37.0 °C, 37.5 °C; blood pressure 120/80 and 135/90; shoe sizes 40, 41, 43. When she divided the total temperature by 3 on her calculator, the answer ended neatly. But when she used the Pythagorean theorem to find the diagonal of the 1 m × 1 m floor tile in the waiting room, her calculator showed 1.41421356… and kept going. She was puzzled.' 2. Say: 'Before we study anything new, write down YOUR weight, height, and shoe size. These are YOUR numbers. Treat them as data — just like Sister Wanjiku's patient data.' 3. Prompt: 'Now try to classify each of your three numbers: Is it odd or even? Prime or composite? And here is the big question — does it end neatly, or could it go on forever?' Allow 4 minutes of silent individual work. 4. Say: 'In your exercise book, write ONE sentence starting with: I predict the calculator shows a never-ending decimal for the tile diagonal because…' Allow 2 minutes. DO NOT confirm or deny predictions at this stage. 5. WAIT TIME: After issuing each prompt, wait a full 12 seconds in silence before scanning the room — this signals that thinking time is genuine.	meaningful quantities. Writing a prediction before instruction activates prior knowledge and creates cognitive dissonance that the rest of the sub-strand will resolve.	mathematical language (e.g., 'fraction', 'divide', 'remainder')? Note 3–4 learners who show misconceptions (e.g., 'all decimals go on forever') for targeted questioning in the Explain phase.
Observe Phase 📹 **VIDEO:** Prime numbers Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos	The teacher writes on the board — or displays on a screen — two side-by-side calculator outputs: (A) (36.8 + 37.0 + 37.5) ÷ 3 = 37.1 (terminates), and (B) $\sqrt{1^2 + 1^2}$ = $\sqrt{2}$ = 1.41421356237… (non-terminating, displayed to 11 decimal places). Learners copy both outputs into a two-column table in their exercise books titled	1. Write both outputs on the board and say: 'These are real outputs from a basic Casio calculator — exactly what Sister Wanjiku would have seen at the health centre.' 2. Say: 'Copy this two-column table: Left column — Numbers That End Neatly. Right column — Numbers That Seem To Go On Forever. Place each number from	Two-Column Observation Table (Claim–Evidence scaffold) — sorting observable evidence (calculator outputs) into two categories before formal definitions are introduced mirrors the NGSS practice of Analysing and Interpreting Data. It also surfaces the key disciplinary idea: some numbers can be expressed	Check that learners can write at least three of the six scenario numbers as fractions with integer numerator and denominator. A learner who writes 36.8 = 368/10 or 46/5 is demonstrating readiness for the formal definition of rational numbers. Flag learners who write $\sqrt{2}$ = 141421356/100000000 (a common misconception — that

📖 **HTML:** [Order of Real Numbers (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	'Sister Wanjiku's Numbers'. They then examine the six numbers from the scenario (36.8, 37.0, 37.5, 120, 80, 40) and try to express each as a fraction (e.g., 36.8 = 368/10). They record which numbers they can write as a fraction and which they cannot, using a simple tick/cross system.	the scenario in the correct column.' 3. After 3 minutes, ask: 'Now try to write each number in the LEFT column as a fraction — a ratio of two whole numbers. For example, can 37.1 be written as a fraction? Discuss with the person next to you for 90 seconds.' 4. WAIT TIME: After the pair discussion, wait 10 seconds before asking for responses. 5. Cold-call 2 learners (by name, not by raised hand): 'Kamau, show us how you wrote 37.0 as a fraction.' Then: 'Achieng, did you write 120 as a fraction? How?' 6. Do NOT yet attempt √2. Say: 'We will come back to the right-hand column — that is the mystery we are here to solve over the coming lessons.'	as a ratio of integers; others apparently cannot.	truncating gives an exact fraction) for a whole-class clarification in the Explain phase.
Explain Phase 📹 **VIDEO:** [Prime numbers] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Order of Real Numbers (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	The teacher introduces the formal vocabulary: rational number (any number expressible as p/q where p and q are integers, q ≠ 0) and irrational number (cannot be expressed as p/q). Learners revisit their two-column table from the Observe phase and relabel the columns. They then classify a new Kenya-context set of eight numbers — 0.75 litres of water in a KIWASCO measure, 3.142 (π approximation on a road-marking stencil), √9, √2, −4, 0, 2/7, and a maize flour price of KSh 55.50 per kg — writing a one-sentence justification for each classification in their exercise books.	1. Write on the board: 'A RATIONAL NUMBER is any number that can be written as p/q, where p and q are integers and q ≠ 0.' Underline 'integers' and 'q ≠ 0'. Say: 'Every number in Sister Wanjiku's left column is rational. Let us prove it together.' 2. Walk through 37.1 = 371/10 and 120 = 120/1 as worked examples, narrating each step. 3. Say: 'Now here is a new list — eight numbers from everyday Kenyan life. For each one, write: RATIONAL or IRRATIONAL, and one sentence explaining why.' Display the eight numbers. Allow 5 minutes of individual work. 4. WAIT TIME: After the 5 minutes, pause for 12 seconds before initiating cold-calls — this gives learners time to review their answers. 5. Cold-call 4 learners (different	Justified Classification with Return to Phenomenon — requiring a written justification for each number (not just a label) engages the NGSS practice of Constructing Explanations. Returning explicitly to the tile diagonal after the formal definition is introduced reinforces the phenomenon as the reason these definitions matter, preventing the lesson from feeling like abstract vocabulary work.	Collect or photograph three learners' exercise books at the end of this phase. Success indicator: the learner correctly classifies at least 6 of 8 numbers AND gives a justification that references the p/q definition (not merely 'it has a decimal' or 'it looks weird'). Pay particular attention to √9 — learners who say 'irrational because it has a square root sign' reveal a critical misconception to address at the start of Lesson 2.

		learners from the Observe phase): (i) 'Njeri, is 2/7 rational? Tell us why.' (ii) 'Otieno, what about √9 — rational or irrational? Surprise us.' (iii) 'Fatuma, the price KSh 55.50 — rational or irrational?' (iv) 'Baraka, what about π ≈ 3.142 written on a road-marking stencil — and does the approximation change anything?' 6. After each response, ask the class: 'Do you agree or disagree? Thumbs up or thumbs sideways.' Address any disagreement immediately. 7. Return to the phenomenon: 'So which column does √2 — the tile diagonal — belong in? Why can we NOT write it as p/q? We will prove this in Lesson 3, but for now, mark it IRRATIONAL and put a question mark next to it on your DQB.'		
Driving Question Board (DQB) Creation 📹 **VIDEO:** Prime numbers Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Order of Real Numbers (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Each learner writes two sticky notes (or two entries in a designated DQB section of their exercise book if physical sticky notes are unavailable): (a) one 'What I Notice' statement and (b) one 'What I Wonder' question, both explicitly connected to Sister Wanjiku's scenario. The teacher organises contributions on the class DQB under three columns: NOTICE | WONDER | NOW I KNOW. Four learners are cold-called to read their Wonder questions aloud; these become the public driving questions the class will return to across the sub-strand.	1. Distribute sticky notes (or direct learners to the DQB page in their books). Say: 'Write ONE thing you noticed from today's lesson on a yellow note — start with: I notice that… Write ONE question you still have on a pink note — start with: I wonder why…' 2. Allow 3 minutes. Circulate and prompt any learner who is stuck: 'Think about Sister Wanjiku's tile diagonal. What surprised you?' 3. Collect notes and rapidly sort them into the three DQB columns, narrating as you go: 'Kamau notices that all the health-centre measurements can be written as fractions — that goes in NOTICE. Achieng wonders why you cannot write √2 as a fraction no matter how hard you try — that goes in WONDER. That is the question that drives our whole investigation.'	Notice–Wonder DQB Launch — this strategy, adapted from the Mathematical Practice of Making Sense of Problems, externalises learners' thinking and creates a publicly shared record of the class's growing understanding. The three-column structure (NOTICE | WONDER | NOW I KNOW) makes intellectual progress visible and gives each subsequent lesson a purpose: to move something from WONDER to NOW I KNOW.	Review all Wonder questions (sticky notes or book entries) after class. Tally: How many learners asked questions about WHY √2 is non-terminating (conceptual)? How many asked HOW to tell if any given number is irrational (procedural)? How many asked WHAT other irrational numbers exist (extensional)? This tally informs the sequencing emphasis in Lessons 2–4.

		4. Cold-call 4 learners to read their Wonder questions aloud. After each, say: 'Hold that question — we will answer it before the end of this sub-strand.' Write the four questions on the board in a box labelled 'Our Driving Questions.' 5. WAIT TIME: After each cold-call reading, wait 10 seconds and ask: 'Who else had a similar wonder? Raise your hand.' This validates diverse thinkers. 6. Point to the DQB and say: 'Every lesson, we will move questions from WONDER to NOW I KNOW. Today, we opened the board. By the end of Sub-Strand 1.1, it should be full.'		
Model Building Phase 📹 **VIDEO:** [Prime numbers] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Order of Real Numbers (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Learners draw a first-draft Number Classification Diagram in their exercise books — a simple two-branching tree or Venn diagram starting from 'Real Numbers' and splitting into 'Rational' and 'Irrational'. They place Sister Wanjiku's six scenario numbers and their own three personal measurements (weight, height, shoe size) into the correct branches, using today's definitions. They annotate the diagram with one open question (a question mark bubble) on the Irrational branch, acknowledging that √2's non-terminating nature is not yet fully explained.	1. Say: 'Scientists and mathematicians build models to organise their thinking. Today you will build your FIRST model of the real number system — it will be rough, and that is fine. We will revise it every lesson.' 2. Draw the scaffold on the board: a box labelled REAL NUMBERS, with two branches drawn but not labelled. Say: 'Label the two branches using what you learned today. Place at least six numbers from Sister Wanjiku's clinic and your own three measurements onto the correct branch.' 3. Allow 4 minutes. Circulate and look for learners placing negative integers (e.g., −4 from the Explain phase) — prompt: 'Which branch does −4 belong on? Can you write −4 as a fraction?' 4. Say: 'On the Irrational branch, draw a question-mark bubble and write: WHY can't √2 be written as p/q? We will fill this bubble in Lesson 3.' 5. WAIT TIME: Before closing, wait	Cumulative Model Diagram (First Draft) — drawing a visual model at the END of the first lesson, before full understanding is established, is a deliberate NGSS Developing and Using Models move. The explicit incompleteness (the question-mark bubble) signals to learners that models are provisional and will be revised — building the epistemic habit of treating knowledge as a work in progress.	At lesson end, do a quick gallery scan: hold up your diagram. Look for three indicators: (1) Two clearly labelled branches — Rational and Irrational. (2) At least four numbers correctly placed. (3) A question-mark bubble present on the Irrational branch. Any learner missing all three needs a brief 2-minute one-on-one check before the next lesson.

		12 seconds and ask: 'Compare your diagram with your neighbour's. Find ONE difference. Is one of you right, or are both valid?' Allow 60 seconds of pair comparison. 6. Close by returning to the phenomenon: 'Sister Wanjiku's tile diagonal lives on the Irrational branch. Every lesson, we will understand that branch better — until we can fully explain why her calculator never stopped.'		

D. TEACHER REFLECTION

After the lesson, ask yourself: (1) When I cold-called the four learners during the DQB phase, did their Wonder questions reveal conceptual thinking (WHY questions) or only procedural questions (HOW TO questions)? If mostly procedural, plan a brief whole-class discussion at the start of Lesson 2 to elevate conceptual wondering. (2) Did any learner claim that $\sqrt{9}$ is irrational because 'it has a square root sign'? If yes, this is a priority misconception — open Lesson 2 with a deliberate counter-example before moving to operations on rational numbers. (3) Were learners emotionally safe when recording personal weight and height? If any discomfort was visible, transition to using fictional patient data (e.g., 'Patient A weighs 58 kg') in future lessons while keeping the Mathare North scenario intact.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Sister Wanjiku's calculator showed $(36.8 + 37.0 + 37.5) \div 3 = 37.1$ (a decimal that terminates), but $\sqrt{(1^2+1^2)} = \sqrt{2} = 1.41421356\ldots$ (a decimal that never ends and never repeats). All six health-centre measurements could be written as fractions with integer numerators and denominators.
What did I learn?	A rational number is any number that can be written as p/q where p and q are integers and $q \neq 0$. All of Sister Wanjiku's health-centre measurements are rational. $\sqrt{2}$ — the tile diagonal — appears to be irrational because no fraction p/q exactly equals it, though we have not yet proved why.
How does this explain the phenomenon?	The Real Number system branches into Rational numbers (which include integers, fractions, and terminating or repeating decimals) and Irrational numbers (which cannot be expressed as p/q and whose decimals neither terminate nor repeat). Sister Wanjiku's average temperature is rational; the waiting-room tile diagonal is irrational — and understanding WHY is the driving question for this sub-strand.

LESSON 2 (40 minutes): Observe: Rational vs. Irrational Numbers — Sorting Kenyan Real-Life Data	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Learners gather evidence by examining a curated set of Kenyan real-life numbers and attempting to express each as a fraction p/q, thereby distinguishing rational from irrational numbers and advancing their explanation of the anchoring phenomenon.
Knowledge	Classify real numbers as rational or irrational.
Skills	Critical thinking and problem solving: the learner develops interpretation and inference skills as they classify numbers as either rational or irrational numbers.
Attitudes	Self-efficacy: the learner develops self-awareness as they discuss the use of real numbers in real life such as in body temperature and blood pressure.
Key Inquiry Question	Why are numbers important?
Purpose in Storyline	In Lesson 1 (Predict), learners made initial guesses about why the health volunteer's calculator display 'ended' for temperature averages but 'went on forever' for √2. In this Observe lesson, learners move from guessing to evidence gathering: they work through a structured Kenyan data set — maize prices, athletics times, tile diagonals, blood pressure values — and systematically test each number for the fraction p/q property. The class sort that follows produces the first shared, evidence-backed distinction between rational and irrational numbers, directly feeding the Driving Question Board and the cumulative class model.
Safety Notes	No physical hazards. Remind learners to handle calculators with care (KICD Value: Responsibility). Emotional sensitivity: when using body-measurement examples (weight, height), frame all numbers as data points, not personal judgements.

B. LESSON OVERVIEW
Learners examine a curated data card of ten Kenyan real-life numbers drawn from health, agriculture, athletics, and construction contexts. Working in pairs, they attempt to write each number as a fraction p/q (with p and q integers, q ≠ 0). Numbers that resist this form are tentatively labelled 'irrational'. The teacher then facilitates a whole-class board sort using cold-calling (minimum 5 cold-calls, 60 seconds silent think each). Pairs update their Driving Question Board sticky notes with new evidence, and the lesson closes with a brief individual exit card that feeds directly into Lesson 3 (Explain).

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** [Prime numbers] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos	Each learner silently re-reads the phenomenon statement displayed on the board: 'The health volunteer's average temperature divided neatly; √2 did not. Why?' They then individually write one sentence completing the stem: 'I	1. Display the phenomenon on the board and read it aloud. Say: 'Before we look at any new numbers today, I want you to silently revisit your prediction from last lesson.' (10 seconds pause.) 2. Distribute the sentence-stem	Prediction journaling — activates prior knowledge and creates cognitive dissonance that motivates evidence gathering. By committing predictions to paper before observing data, learners are primed to notice when	Teacher scans written sentences during the 60-second window. Observable indicator: learner has written a complete sentence referencing either fractions, 'exact' decimals, or division. Learners who write 'I don't know' are noted

📖 **HTML:** [Properties of Rational Numbers (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	think the temperature average ends neatly because ___.' They do NOT share yet — this is silent prediction to activate prior thinking from Lesson 1 and create a knowledge gap that the observation activity will fill.	card or write on board: 'I think the temperature average ends neatly because ___.' 3. Allow 60 seconds of silent individual writing — enforce silence with a visible countdown timer. 4. Cold-call 2 learners (select learners who have NOT spoken yet in the unit): 'Achieng, read your sentence to us.' Then: 'Mutua, does your prediction agree or differ? Tell us one word.' 5. Say: 'Hold those ideas. Today we gather evidence to test them.' Do NOT confirm or deny any prediction.	evidence confirms or contradicts their thinking.	for targeted support during the pair activity.
Observe Phase 📹 **VIDEO:** Prime numbers Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Properties of Rational Numbers (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Pairs receive a printed or board-displayed 'Kenyan Numbers Data Card' containing ten numbers drawn from authentic Kenyan contexts: 1. Average maize price at Wakulima Market, Nairobi: KSh 42.50 per kg 2. Eliud Kipchoge's average pace in a training run: 3.5 minutes per km 3. Diagonal of a 1 m × 1 m EACC standard floor tile: √2 m ≈ 1.41421356… 4. Blood pressure reading: 120/80 (the ratio as a fraction) 5. Body temperature recorded at Mathare North Health Centre: 37.5 °C 6. Circumference-to-diameter ratio of a circular ugali plate: π ≈ 3.14159265… 7. A farmer in Kericho's land fraction sold: 3/7 of a hectare 8. Diagonal of a 1 m × 2 m rectangular tile: √5 m ≈ 2.2360679… 9. Athletics track lane width at Moi International Sports Centre: 1.22 m	1. Distribute Data Cards (or uncover the board list). Say: 'With your partner, your job is to be mathematical detectives. For each number, ask: can I write this EXACTLY as a fraction — an integer divided by a non-zero integer? If yes, it is rational. If no matter what you try, you cannot — it is irrational.' 2. Model ONE example aloud: 'Look at number 2 — 3.5 minutes per km. Can I write 3.5 as a fraction? Yes: 3.5 = 7/2. So it is rational. Write R.' 3. Set a 12-minute pair-work timer. Circulate — do NOT give answers. Prompt stuck pairs with: 'Try writing the decimal as a fraction. What happens when you divide on your calculator?' 4. For pairs who finish early, extension prompt: 'Can you find ONE more number from your life this morning — before school — that is rational? One that might be irrational?' 5. At the 10-minute mark, give a 2-minute warning: 'Wrap up — you need at least a label and one sentence for each number.'	Data Card Sort with fraction-testing protocol — learners use a concrete operational procedure (attempt p/q representation) as an evidence-gathering tool. This grounds the abstract rational/irrational distinction in repeated hands-on testing, connecting each number back to a real Kenyan context, and mirrors how the health volunteer experienced the phenomenon.	Teacher circulates and checks three specific pairs' cards mid-activity. Observable indicators: (a) learner correctly identifies at least 6 of 10 numbers; (b) justification sentences reference the fraction form (e.g., 'I could write it as 37.5/1 = 75/2'); (c) learners correctly flag √2 and π as irrational. Red flag: pairs labelling all numbers 'rational' — prompt with 'Type √2 into your calculator. Can you write all those digits exactly as a fraction?'

	10. Githeri protein content approximation: 0.̇3 (i.e., 0.333… = 1/3) For EACH number, pairs must: (a) attempt to write it as p/q; (b) mark it R (rational) or I (irrational) on their card; (c) write one justification sentence. They use calculators where needed.			
Explain Phase 📹 **VIDEO:** Prime numbers Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Properties of Rational Numbers (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	The teacher draws a large two-column table on the board: RATIONAL | IRRATIONAL. Learners are cold-called one pair at a time to place a number and defend their choice. After all ten are placed, learners write a class definition of each term in their exercise books in their own words, then compare with a partner. They then return to the phenomenon: 'Which of the health volunteer's numbers are rational? Which is irrational? How does this help us answer the driving question?'	1. Draw the two-column board table. Say: 'We are going to build our class evidence board. I will cold-call pairs — you must state the number, your label, AND your reason. No hands — I choose.' 2. Cold-call minimum 5 pairs (count aloud: 'That is cold-call number one… number two…'). For EACH cold-call: (a) name the pair; (b) allow 60 seconds of silent think (visible countdown); (c) ask for label and justification; (d) ask the class: 'Thumbs up if you agree, thumbs sideways if you are unsure, thumbs down if you disagree — hold until I say go.' (e) Allow 1 challenger if a thumb-down is raised: 'Wanjiku, you disagree — give us your evidence.' 3. After all 10 are placed, ask: 'Look at the irrational column. What do these decimals have in common?' (Wait 60 seconds silent think.) Cold-call 1 learner. Expected answer: they are non-terminating and non-repeating. Write this on the board as a class generalisation. 4. Connect to phenomenon: 'So back to Mathare North — which of the volunteer's numbers are rational? Point to your card.' Cold-call 1 learner for temperature average, 1 for √2. Ask: 'Why did the calculator show a neat ending for temperature but not for √2?' 5.	Claim-Evidence-Reasoning (CER) Board Sort — each cold-called learner must state a claim (R or I), cite evidence (the fraction attempt or calculator output), and give reasoning (why it does or does not satisfy p/q). The thumbs protocol gives all 40+ learners an active response role simultaneously, preventing passive observation. Connecting back to the phenomenon at the end of the phase anchors the new vocabulary in the original puzzling context.	Observable indicators during cold-calls: learner uses the phrase 'I can/cannot write it as p/q' in their justification. Teacher records on a clipboard which learners gave correct CER responses (target: 5 different learners across 5 cold-calls). Written definitions checked: does the learner's definition include (a) integers p and q, (b) q ≠ 0, and (c) a contrast with irrational (non-terminating, non-repeating)? Definitions that omit q ≠ 0 are flagged for Lesson 3 follow-up.

		Learners write their own definition of rational and irrational in exercise books (2 minutes). Pair-share: 'Read your definition to your partner. Agree on one best version.'		
Driving Question Board (DQB) Creation 📹 **VIDEO:** Prime numbers Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Properties of Rational Numbers (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Each learner receives two sticky notes (or writes in a designated DQB section of their notebook if physical notes are unavailable). On the YELLOW note they write one thing the Observe activity proved or confirmed ('I now know…'). On the BLUE note they write one new question or remaining confusion raised by the data ('I still wonder…'). Pairs share before posting. The teacher reads 3–4 blue notes aloud to the class, framing them as questions the sub-strand will keep investigating.	1. Say: 'Our Driving Question Board is our shared evidence wall for the whole sub-strand. Every lesson we add to it.' 2. Distribute sticky notes or direct learners to their DQB notebook section. Say: 'Yellow note: what does today's evidence PROVE about the health volunteer's puzzle? Blue note: what do you still wonder — what does our class evidence NOT yet explain?' 3. Allow 3 minutes for individual writing. 4. Pairs share notes before posting — 'Tell your partner what you wrote. If they wrote something better, use theirs.' 5. Teacher selects and reads aloud 3 blue notes that represent high-leverage questions (e.g., 'Why exactly does √2 go on forever?', 'Are there more irrational numbers than rational ones?', 'Can irrational numbers be used in calculations?'). Say: 'These are excellent questions. Keep them — we will answer them in Lessons 3 and beyond.' 6. Post all notes visibly.	Driving Question Board update — making thinking visible and cumulative. The two-colour protocol separates established evidence (yellow) from productive uncertainty (blue), modelling the scientific practice of distinguishing what is known from what remains to be investigated. Reading blue notes aloud validates curiosity and previews Lesson 3 content.	Teacher reads all yellow notes before the end of class. Observable indicator: yellow note references rational/irrational distinction in connection to the phenomenon (e.g., 'I now know the temperature average ends because it is rational — it can be written as 111/3 = 37'). Blue notes that reveal misconceptions (e.g., 'I wonder if π could be a fraction if we use more digits') are documented for targeted addressing in Lesson 3.
Model Building Phase 📹 **VIDEO:** Prime numbers Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Properties of	Learners return to the class model begun in Lesson 1 (a sketch of the number system drawn on the board or in notebooks). Using the Observe evidence from today, each learner adds to their individual model: they draw two clearly labelled branches — RATIONAL and IRRATIONAL — under REAL NUMBERS, and place at least two Kenyan examples from today's data card in	1. Say: 'In Lesson 1 you drew a first sketch of real numbers. Today we upgrade it with evidence.' Display the Lesson 1 class model (if drawn on board) or instruct learners to open to their Lesson 1 model page. 2. Say: 'Add two branches: RATIONAL and IRRATIONAL. Place at least two examples from today's data card in each branch. For √2, write an annotation explaining WHY it is	Progressive model revision — learners maintain a cumulative, evolving representation of the real number system across all 8 lessons. Revising the model forces learners to integrate new evidence with prior understanding rather than treating each lesson as isolated. Comparing models with a partner surfaces alternative conceptions and promotes self-correction before misconceptions	Teacher circulates during model-building and checks that: (a) the rational/irrational branching is correctly labelled; (b) Kenyan examples are placed in the correct branch; (c) √2 annotation references 'non-terminating, non-repeating'. Exit check: teacher asks the cold-called pair — 'If the health volunteer found a new number, 0.$\bar{6}$, which branch? Why?' Observable indicator: learner says

Rational Numbers (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	each branch. They annotate √2 with the note: 'non-terminating, non-repeating — this is why the health volunteer's calculator never stopped.' Pairs compare models and agree on any corrections before the lesson ends.	irrational — use today's evidence.' 3. Allow 4 minutes individual model-building. 4. Pair-compare: 'Show your partner your model. Do your examples match? Did you both annotate √2 the same way?' (2 minutes.) 5. Cold-call 1 pair to share their model with the class. Ask: 'How is your model today different from your Lesson 1 sketch? What changed?' 6. Preview: 'Next lesson we will EXPLAIN why — mathematically — rational decimals must terminate or repeat, and irrational ones cannot. Your model will grow again.'	solidify.	'rational — it equals 2/3' without hesitation. Learners who cannot answer are provided a written prompt card to review before Lesson 3.

D. TEACHER REFLECTION

After the lesson, reflect on the following questions: (1) Did cold-called learners consistently connect their R/I label to the p/q definition, or did they rely only on 'it looks like it goes on forever'? If not, plan an explicit counter-example for Lesson 3 opener (0.333… looks like it goes on forever yet is rational = 1/3). (2) Were there gender or participation patterns in cold-calling? Review your clipboard tally — aim for gender balance across the 5 cold-calls per lesson. (3) Which blue DQB notes reveal the highest-leverage misconceptions to address in Lesson 3? (4) Did the 60-second wait time feel uncomfortable — did you reduce it? Commit to the full 60 seconds: research shows rural and urban Kenyan classrooms both benefit from extended wait time for mathematical justification tasks. (5) Were all pairs able to access the data card? If reading level was a barrier, consider reading each number aloud with its context in Lesson 3 as a scaffold.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners examined a curated set of ten Kenyan real-life numbers (maize prices, athletics times, tile diagonals, blood pressure values, body temperatures, π) and attempted to express each as a fraction p/q.
What did I learn?	Numbers that can be written as p/q (where p and q are integers and q ≠ 0) are rational; numbers such as √2 and π — which produce non-terminating, non-repeating decimals — cannot be written this way and are irrational. The health volunteer's temperature average is rational; the tile diagonal (√2) is irrational.
How does this explain the phenomenon?	The class constructed a Claim-Evidence-Reasoning board sort and revised their real number system model to show two branches (rational and irrational) with Kenyan examples in each, directly connecting back to why the health volunteer's calculator 'ended neatly' for one calculation but 'went on forever' for another.

LESSON 3 (40 minutes): Explain: Defining and Formalising Rational vs. Irrational Numbers	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Learners formalise the mathematical definitions of rational and irrational numbers, correctly classify a range of Kenyan-context real numbers, and update the Driving Question Board with precise mathematical language — deepening their ability to explain why the health volunteer's numbers behave differently.
Knowledge	A rational number is any number that can be expressed in the form p/q where p and q are integers and q ≠ 0. An irrational number cannot be expressed in this form; its decimal expansion is non-terminating and non-repeating. Together, rational and irrational numbers form the set of real numbers.
Skills	Classify real numbers as rational or irrational; justify classification with mathematical reasoning; place rational and irrational numbers on a number line; distinguish between an irrational number and its rational approximation (e.g. π vs. 22/7).
Attitudes	Appreciate that precise mathematical definitions — not just calculator displays — are the reliable tool for deciding whether a number 'ends neatly'; develop confidence in applying formal definitions to everyday Kenyan contexts.
Key Inquiry Question	Why are numbers important?
Purpose in Storyline	In Lessons 1–2, learners gathered evidence that some numbers from the health centre (temperatures, blood pressures, shoe sizes) terminate or repeat while others (√2, π) do not. Lesson 3 gives learners the formal mathematical language — rational vs. irrational — to explain that evidence precisely. By the end of this lesson learners can articulate WHY 37.5 °C 'ends neatly' (it equals 75/2, a ratio of integers) while √2 does not (it cannot be written as any ratio of integers), directly answering the driving question at a deeper level.
Safety Notes	No physical hazards. Remind learners to handle calculators and any printed classification cards with care. Ensure respectful discussion when peers share different classifications — mathematical disagreement must be expressed constructively.

B. LESSON OVERVIEW

Learners transition from evidence-gathering to formal sensemaking. The teacher uses the health-centre phenomenon as the anchor to introduce and model the definitions of rational (p/q, q ≠ 0) and irrational numbers, placing √2 and 22/7 on the number line and explicitly distinguishing 22/7 from π. Learners then apply the formal definitions to classify 20 Kenyan-context numbers on a structured card-sort activity, defend their classifications to peers, and update the class Driving Question Board with precise vocabulary. The lesson closes with a formative exit-ticket tied directly to the phenomenon.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** Prime numbers Source: Khan Academy (English	Learners receive a mini-whiteboard prompt (or jotter): 'Before we define anything formally, sort these five numbers into two groups — Group A:	Display the five numbers on the board beside the health-centre scenario from Lesson 1 (temperatures, √2 tile diagonal). Say: 'We have been collecting	Elicit-then-Confront: surfacing prior conceptions before instruction ensures that the formal definition directly addresses — and visibly corrects — the most common	Teacher observes mini-whiteboard groupings during circulation. Look for: (1) Does the learner recognise −3 as a ratio (−3/1)? (2) Does the learner incorrectly equate 22/7

- US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Properties of Rational Numbers (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	numbers you think CAN be written as a fraction of two whole numbers; Group B: numbers you think CANNOT. Numbers: 0.75, √5, −3, 22/7, π. Write your groupings and one sentence explaining your reasoning for at least one choice.' Learners work silently and individually for 3 minutes, then share with a shoulder partner for 1 minute.	evidence about numbers that end neatly and numbers that go on forever. Today we build the formal mathematical tools to explain WHY. Before I define anything, I want to know what YOU already think.' Circulate silently during individual think time — do NOT correct errors yet. After shoulder-partner sharing, cold-call 3 learners (choose one who has all correct, one who has a common misconception about 22/7, one who is unsure about −3). Provide 12 seconds of WAIT TIME after each cold-call before accepting an answer. Record each learner's grouping on the board without evaluating: 'Thank you — let's hold that idea and test it against the formal definition.'	misconception (that 22/7 IS π, and that negative integers cannot be rational). This creates cognitive readiness for the definition.	with π? (3) Does the learner correctly flag √5 as non-fractional? These three indicators reveal the conceptual gaps the Explain phase must address.
Observe Phase 📹 **VIDEO:** Prime numbers Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Properties of Rational Numbers (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners observe a structured teacher-led worked demonstration (the 'model' is the teacher's written and spoken reasoning, not a physical experiment). Part A — Teacher writes on the board and reads aloud the formal definition: 'A rational number is any number that can be expressed as p/q where p and q are integers and q ≠ 0.' Teacher then maps each health-centre data point onto the definition: 36.8 = 368/10 = 184/5 ✓; 37.0 = 37/1 ✓; 37.5 = 75/2 ✓; shoe size 41 = 41/1 ✓; blood pressure numerator 120 = 120/1 ✓. Learners copy each mapping into their notebooks. Part B — Teacher draws a number line from 1 to 2 on the board, marks 1 and 2, then uses the Pythagorean construction narrative ('a square floor tile of side 1 m has a diagonal of √2 m — Euclid proved this cannot be written as p/q') to place √2 ≈ 1.414 between 1.41 and 1.42.	Use a two-colour chalk or marker system throughout: BLUE for rational, RED for irrational. Say explicitly: 'Every time I write a number in blue it means I can prove it equals p/q. Every time I write in red, I cannot.' When placing √2, say: 'My calculator shows 1.41421356… Notice the digits do not repeat in any pattern — this is our evidence from Lesson 2. Today the definition confirms: no fraction of integers equals √2 exactly.' Pause for 10 seconds of silent processing after placing each number on the number line. Cold-call 2 learners to narrate back: 'Can you tell the class in your own words why I wrote 37.5 in blue?' and 'Why did I write π in red but 22/7 in blue — aren't they almost the same number?' Provide 12 seconds WAIT TIME before accepting responses. Correct the 22/7 vs. π confusion explicitly: 'They are	Colour-coded Worked Example with Explicit Narration: the two-colour system creates a persistent visual anchor that learners can reference throughout the classification activity. Narrating each step aloud models the expert reasoning process (mathematical proof-by-definition) that learners will apply independently.	After the 22/7 vs. π moment, teacher poses a quick show-of-hands check: 'Thumbs up if you now believe 22/7 and π are different kinds of numbers; thumbs sideways if you are still unsure.' Count the sideways thumbs — if more than 4 learners are unsure, re-explain using the decimal expansion of 22/7 (= 3.142857142857… repeating block of six digits — RATIONAL) versus π (3.14159265… no repeating block — IRRATIONAL) before proceeding.

	Teacher then places 22/7 ≈ 2.857 on a second number line and π ≈ 3.14159… beside it, explicitly writing: '22/7 = p/q → RATIONAL approximation; π = non-terminating, non-repeating → IRRATIONAL. They are NOT equal.' Learners annotate their number lines with these labels.	close in VALUE but completely different in NATURE — one is rational, one is irrational. This is why the definition matters more than the calculator display.'		
Explain Phase 📹 **VIDEO:** [Prime numbers] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Properties of Rational Numbers (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Learners complete the '20 Kenyan Numbers' classification card-sort. Each pair of learners receives (or copies from the board) a table of 20 numbers drawn from Kenyan everyday life. They must: (a) write R (rational) or I (irrational) beside each number; (b) for every R-classification, write the number in p/q form with integers p and q; (c) for every I-classification, write one sentence explaining why no such fraction exists. The 20 numbers include: body temperature 36.8 °C; price of unga (maize flour) per kg expressed as 58.50 KES; shoe size 42; √3 (diagonal of a githeri pot's rectangular base); 0.333… (1/3 of a litre of milk); −7 (temperature drop in °C at Nanyuki at night); 22/7; π; √4; 0.125 (1/8 kg of sugar); √7; 2.6̅ (2.666…); the population ratio of Kisumu to Mombasa expressed as 1,155,574/1,208,333; √9; 0; −2/5; √2; 5.0; √16; and 0.101001000100001… (a constructed non-repeating decimal). After 10 minutes of pair work, pairs share answers with a second pair ('pair-to-pair') for 3 minutes, resolving disagreements by citing the definition.	Distribute or display the 20-number table. Say: 'Use ONLY the formal definition we just wrote — p/q, q ≠ 0, p and q integers — not your calculator, not your feelings about the number. The definition is your authority.' Circulate actively. Target three common sticking points: (1) √4 = 2 = 2/1 → RATIONAL — many learners will reflexively mark all surds as irrational; prompt with 'Can you simplify √4 first?'; (2) 0 = 0/1 → RATIONAL; (3) 0.101001000100001… — the pattern grows but never repeats a fixed block, so it is IRRATIONAL. After pair-to-pair sharing, cold-call 4 pairs to report one number each (choose pairs who handled √4, π vs. 22/7, 0, and the constructed decimal). Provide 10 seconds WAIT TIME after each question. After each report, address the class: 'Does every group agree? If you classified differently, stand by your answer and give your p/q evidence.'	Card-Sort with Justification-Writing: requiring learners to write p/q or a sentence for every classification prevents surface-level guessing and forces engagement with the definition. The pair-to-pair protocol surfaces disagreements safely and builds mathematical argumentation — an NGSS Science and Engineering Practice (Constructing Explanations and Engaging in Argument from Evidence).	Collect or photograph each pair's completed classification table. Success criteria: (a) 17–20 correct with valid p/q or sentence justifications = Meets/Exceeds Expectations; (b) 12–16 correct = Approaches Expectations (common errors on √4, √9, √16 and constructed decimal); (c) fewer than 12 = Below Expectations — flag for targeted re-teaching in Lesson 4. Track specifically whether learners correctly identify √4, √9, and √16 as rational — this is the most diagnostic misconception for this lesson.
Driving Question Board (DQB) Creation	Learners return to the class DQB (established in Lesson 1). Working in their pairs, they write two sticky notes or index cards: Card 1	Point to the DQB and say: 'In Lesson 1 we asked WHY some of the health volunteer's numbers end neatly. In Lesson 2 we	DQB Progressive Formalisation: the DQB tracks the class's growing understanding across the 8-lesson sequence. Adding a 'Formalised	Teacher reads all green cards during the gallery walk. Acceptable card: must include p/q, must state q ≠ 0, must state p and q are

📹 **VIDEO:** [Prime numbers](#) Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Properties of Rational Numbers (Flexbook)](#) Source: CK-12 🔍 Search ARES for similar readings	(GREEN — 'What we now KNOW formally'): a statement using the words 'rational,' 'p/q,' 'integers,' and 'irrational.' Card 2 (ORANGE — 'New question this lesson raised'): a genuine question the lesson has left open for them. Pairs post their cards on the DQB under the columns 'Formalised Understanding' and 'Still Wondering.' The teacher leads a 3-minute gallery walk where learners read three other pairs' green cards and give a silent thumbs-up if the definition is correctly stated.	observed the evidence in decimals. Today we have the formal answer — but I want YOU to write it, not me.' After pairs post their cards, facilitate the gallery walk. Then, as a whole class, select the two best-worded green cards and read them aloud. Say: 'These two cards are now our shared formal definition. Everyone copy them into your notebook under the heading: Lesson 3 — Formal Definition.' Read aloud a selection of orange 'Still Wondering' cards to preview Lessons 4–5 ('How do we do arithmetic WITH irrational numbers?' 'Are there more irrational numbers than rational ones?'). Explicitly connect back to the phenomenon: 'The health volunteer's average temperature — 37.1 °C — is rational. The tile diagonal √2 is irrational. We can now prove both statements using the definition on our DQB.'	Understanding' column in Lesson 3 marks the conceptual shift from 'evidence about patterns' (Lessons 1–2) to 'definition-based proof' — making the epistemic progress visible to learners.	integers, must contrast with irrational. Cards missing any of these four elements are flagged — teacher returns them with a one-sentence prompt: 'Your card is missing one part of the definition — which of the four conditions have you left out?' This is a specific, observable indicator of definitional completeness.
Model Building Phase 📹 **VIDEO:** [Prime numbers](#) Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Properties of Rational Numbers (Flexbook)](#) Source: CK-12 🔍 Search ARES for similar readings	Learners update their personal 'Real Number Model' — a cumulative diagram begun in Lesson 1 and revised in Lesson 2. In Lesson 3 they add: (a) formal labels 'Rational: p/q, q ≠ 0' and 'Irrational: non-terminating, non-repeating' to the two branches of their diagram; (b) a number-line segment between 1 and 2 showing √2 ≈ 1.414… plotted and labelled I (irrational); (c) a second segment showing 22/7 plotted and labelled R (rational), and π plotted nearby labelled I (irrational) with a double-headed arrow and the annotation 'close in value — different in nature'; (d) at least 3 new numbers from the 20-number card sort, colour-coded blue (rational) or red (irrational). Learners write one	Say: 'Your model from Lesson 1 showed two clouds — numbers that end and numbers that don't. Today you have the mathematical skeleton to replace those clouds with precise branches. Add the formal labels now.' Give learners 5 minutes for silent individual model revision. Circulate and look for: Are learners using the two-colour system consistently? Have they placed √2 correctly between 1.41 and 1.42? Have they separated π and 22/7 on their number line? Cold-call 2 learners to share their 'Model Update Statement' aloud. After each, ask the class: 'Does this statement connect back to the health volunteer's data? Give a thumbs-up if yes.' Close the lesson by pointing to the model and	Cumulative Model Revision with Metacognitive Writing: the 'I revised my model today because…' sentence stem is a metacognitive tool that forces learners to articulate conceptual change — linking new formal knowledge to prior evidence and to the anchoring phenomenon. This embodies the NGSS practice of Developing and Using Models.	Collect models at the end of the lesson (or photograph them). Look for three specific indicators: (1) Both formal labels (rational p/q and irrational non-terminating non-repeating) are present — if absent, the learner has not internalised the definition; (2) √2 and π are correctly coloured red, and 22/7 is correctly coloured blue — if 22/7 is red, the 22/7 vs. π misconception persists; (3) The 'Model Update Statement' references the health-centre phenomenon by name or by reference to a specific data point — generic statements indicate surface engagement. These three indicators directly inform differentiation planning for Lesson 4.

	'Model Update Statement' at the bottom of their diagram: 'I revised my model today because…' connecting to the health-centre phenomenon.	saying: 'Every lesson from now on will add more structure to this model. By Lesson 8, you will be able to explain every number the health volunteer recorded — and every number you encounter in Kenyan daily life — using this framework.'		

D. TEACHER REFLECTION

After the lesson, reflect on the following questions in your professional journal:

1. DEFINITIONAL COMPLETENESS: When learners wrote their green DQB cards, how many included all four components of the rational number definition (p/q, p and q integers, q ≠ 0, contrast with irrational)? If fewer than 70% of cards were complete, plan a 5-minute definitional drill to open Lesson 4.

2. THE 22/7 vs. π MISCONCEPTION: Did the thumbs-sideways check reveal persistent confusion between 22/7 and π? If so, which part of the explanation failed — the decimal expansion argument, the definition-based argument, or the number-line placement? Adjust the Lesson 4 warm-up accordingly.

3. CARD-SORT DIAGNOSTICS: Which of the 20 numbers generated the most disagreement between pairs? Were the problem numbers the 'hidden rationals' (√4, √9, √16, 0) or the 'constructed irrational' (0.101001000100001…)? This tells you whether learners are over-applying the surd = irrational rule or whether they struggle with non-repeating decimal logic.

4. WAIT TIME IMPACT: Did cold-called learners give more complete answers after 10–12 seconds of wait time than in previous lessons? Note any learners who consistently needed more than 12 seconds — these learners may benefit from seeing the question written on the board before being cold-called.

5. PHENOMENON CONNECTION: In the Model Update Statements, did learners spontaneously reference the Mathare North Health Centre scenario, or did they write generic statements? Strong phenomenon connection indicates that the storyline thread is working; weak connection suggests you need to re-open the phenomenon explicitly at the start of Lesson 4.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	In Lessons 1 and 2, we observed that the health volunteer's temperature average (37.1 °C) produced a terminating decimal on the calculator, while the tile diagonal (√2) produced a never-ending, non-repeating decimal. We also observed that 22/7 produces a repeating decimal (3.142857) while π produces a non-repeating one.
What did I learn?	In Lesson 3, we learned the formal definitions: a RATIONAL number can always be written as p/q where p and q are integers and q ≠ 0 — its decimal either terminates or repeats. An IRRATIONAL number cannot be written in this form — its decimal is non-terminating and non-repeating. We learned that 22/7 is RATIONAL (it equals a fraction of integers) while π is IRRATIONAL, even though they are close in value.
How does this explain the phenomenon?	Using these definitions, we can now explain the health volunteer's puzzle: her temperature readings (36.8 = 184/5, 37.0 = 37/1, 37.5 = 75/2) are all rational — they are exact ratios of integers, so they terminate on the calculator. The tile diagonal (√2) is irrational — no ratio of integers equals it exactly — so the calculator display never ends. The difference is not about the calculator; it is about the nature of the number itself.

LESSON 4 (40 minutes): Combined Operations on Rational Numbers I — Addition & Subtraction in the Market and Classroom	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Learners explore and apply addition and subtraction of rational numbers — including fractions and decimals — using Kenyan market and school contexts, and begin to ask whether combined operations on rationals always produce a rational result.
Knowledge	– Define combined operations on rational numbers and state the BODMAS rule for ordering addition and subtraction. – Explain why the sum or difference of two rational numbers is always a rational number (closure property). – Identify rational numbers in market prices (decimals) and stationery distribution (fractions) as numbers that can be written in the form p/q.
Skills	– Perform combined addition and subtraction on rational numbers expressed as fractions and decimals using BODMAS. – Represent and solve real-life problems (market costs, stationery distribution) involving addition and subtraction of rational numbers and communicate solutions clearly to peers.
Attitudes	– Appreciate that understanding combined operations on rational numbers leads to more accurate and fair decisions in everyday Kenyan life (market transactions, resource sharing). – Demonstrate respect by listening attentively to peers' solution strategies during group presentations and taking turns constructively.
Key Inquiry Question	Why are numbers important?
Purpose in Storyline	Having formally defined rational and irrational numbers in Lessons 1–3, learners now put rational numbers to work: they perform combined addition and subtraction operations in Kenyan market and school contexts, gathering evidence that combining rationals always yields a rational result — a key step toward answering the driving question about why some numbers 'end neatly'.
Safety Notes	No specific hazards. Ensure group work is conducted respectfully; be sensitive to learners who may find market/financial contexts embarrassing due to personal economic situations — frame all scenarios as community case studies, not personal disclosures.

B. LESSON OVERVIEW

Lesson 4 sits at the heart of the evidence-gathering sequence: learners have classified real numbers (Lessons 1–2) and formalised the rational/irrational distinction (Lesson 3). Now they begin to operate on rational numbers, starting with addition and subtraction. The lesson opens with a prediction task anchored to the sukuma wiki vendor scenario — a familiar Nairobi roadside market stall — asking learners whether the total cost of three food items will 'end neatly' or 'go on forever'. This connects directly to Sister Wanjiku's temperature average from the anchoring phenomenon: that average terminated, and learners will gather evidence for why.\n\nIn the Observe phase, the teacher models BODMAS/PEMDAS with a worked market-price scenario (ugali flour + beans + sukuma wiki), deliberately using both fraction and decimal representations of prices so learners see the same rational number in two forms. Learners then move into groups of three to work through a carefully scaffolded stationery-distribution case scenario set at a Kisumu primary school — each group receives a different version of the problem involving addition and subtraction of fractions and decimals. Each group nominates one spokesperson to present their solution, building the communication and collaboration competency mandated by KICD.\n\nThe Explain phase draws out the closure property of rational numbers under addition and subtraction: if you add or subtract two numbers that can each be written as p/q, the result can also be written as p/q. This is the sensemaking heart of the lesson and directly advances the driving question. The lesson closes with a Driving Question Board update and a model revision in which learners add a 'Rational Numbers: Operations' layer to their cumulative real-number diagram.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** Worked example: Order of operations (PEMDAS) Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Areas of Combined Figures Involving Circles (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners read a short scenario card (displayed on the board or read aloud by the teacher): 'Mama Akinyi runs a roadside stall near Kisumu Bus Station. On Monday morning she sells ugali flour for Ksh 45.50, a bunch of sukuma wiki for Ksh 12.75, and a tin of beans for Ksh 38.00. A customer buys all three. Will the total cost be a number that ends neatly, or will it go on forever like √2?' Individually (no discussion yet), learners write their prediction and a one-sentence justification in their exercise books. They also predict: 'If I add two rational numbers, will the answer always be rational?' Learners share predictions with a shoulder partner for 60 seconds.	Display the scenario on the board and read it aloud with expression. Say: 'Before we calculate anything, I want every one of you to write your own prediction — not your neighbour's. You have 90 seconds. Go.' Set a visible timer. After 90 seconds, say: 'Turn to your shoulder partner and share your prediction. You have 60 seconds.' [WAIT TIME: allow full 60 seconds of partner talk.] Cold-call 3 learners who have NOT yet spoken (scan the room for quiet learners): 'Amina — what did you predict, and why?' / 'Odhiambo — do you agree or disagree with Amina? Give one reason.' / 'Chebet — which part of Lessons 1 to 3 helped you make your prediction?' Record two contrasting predictions on the board under the heading 'OUR PREDICTIONS' — do NOT confirm or correct yet. Say: 'We will come back to these predictions at the end of today's lesson.'	Predict-Justify-Compare: Learners first commit to an individual written prediction (activating prior knowledge about rational number decimal behaviour from Lesson 3), then compare with a partner to surface prior conceptions. Recording contrasting predictions on the board creates productive cognitive tension that the Observe phase will resolve.	Teacher scans exercise books during the 90-second individual writing time, looking for: (1) whether learners connect 'ending neatly' to the definition of a rational number (p/q form) from Lesson 3; (2) whether any learners predict the total will be irrational — this signals a misconception to address during the Explain phase. Note names of learners with correct justifications and those with misconceptions for targeted cold-calling later.
Observe Phase 📹 **VIDEO:** Worked example: Order of operations (PEMDAS) Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Areas of Combined Figures Involving	The teacher works through a BODMAS demonstration on the board using the sukuma wiki market scenario, showing both fraction and decimal forms of the same prices. Learners copy the worked example, annotating each step with the operation name. They then work in pre-assigned groups of three on a stationery-distribution case scenario (Kisumu school context) involving addition and subtraction of rational numbers. Each group has a slightly different set of numbers (fractions, mixed decimals). After	TEACHER DEMONSTRATION (5 minutes): Write on the board: 'Mama Akinyi's total = 45.50 + 12.75 + 38.00'. Say: 'Watch what I do — I am going to show you the same calculation using decimals first, then using fractions, and I want you to notice something important.' Work through the decimal addition carefully, narrating each step aloud: 'I align the decimal points — this is the BODMAS rule in action, addition proceeds left to right when there is no multiplication or division involved.' Then rewrite 45.50 as	Worked Example with Parallel Representations: The teacher deliberately shows the same calculation in decimal and fraction form side by side. This strengthens the Lesson 3 understanding that rational numbers have two equivalent representations, and makes visible the BODMAS structure. The group case scenarios use a Jigsaw-lite presentation structure (each group brings a unique problem to the class) so the full class observes multiple instances of rational-number addition and subtraction,	During group work, teacher listens for: (1) correct use of common denominators when adding fractions; (2) correct alignment of decimal points; (3) students articulating that their answer is still expressible as p/q. During presentations, teacher checks whether the spokesperson can name the form of the answer (rational/irrational). Red flag: any group whose final answer is a non-terminating decimal — investigate whether they made an arithmetic error or a conceptual error about what constitutes a rational number.

Circles (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	10 minutes of group work, one spokesperson per group presents their group's solution at the front, writing it on the board step by step while the group explains the reasoning.	91/2, 12.75 as 51/4, and 38.00 as 38/1, and add them using a common denominator of 4. Say: 'Both methods give Ksh 96.25 — a number that terminates. Checkpoint question — hands down, I am cold-calling: Wanjiru, why does 91/2 + 51/4 + 38 not go on forever like √2?' [WAIT TIME: 12 seconds of silence before accepting the answer.] After Wanjiru responds, say: 'Kamau, can you add one more reason to what Wanjiru said?' GROUP WORK (10 minutes): Distribute scenario cards. Say: 'In your group of three, read the scenario, decide which person writes, which explains, and which checks. Every member must be able to defend every step. You have exactly 10 minutes.' Circulate; crouch to group level; ask probing questions: 'Why did you choose that common denominator?' / 'Is your answer in its simplest form?' / 'Is your answer still a rational number — how do you know?' Do NOT give answers. GROUP PRESENTATIONS (7 minutes): 'Each group sends one spokesperson to the board. Spokesperson, write AND explain — do not just read numbers.' After each presentation, ask the class: 'Thumbs up if your group used a different method to reach the same answer.'	building inductive evidence for the closure property.	
Explain Phase 📹 **VIDEO:** Worked example: Order of operations (PEMDAS) Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos	Learners return to their seats. The teacher guides a whole-class discussion to extract the closure property: 'If a = p/q and b = m/n (both rational), then a + b = (pn + mq)/(qn) — still a fraction of integers, still rational.' Learners write this algebraic argument in their books using a concrete example drawn from their group's	Say: 'Let me show you why the total in every group's scenario had to be rational — not by accident, but by mathematical law.' Write on board: 'Let a = p/q and b = m/n where p, q, m, n are integers and q ≠ 0 and n ≠ 0.' Work through the addition: 'a + b = pn/qn + mq/qn = (pn + mq)/(qn). Is (pn + mq) an integer? YES — products and	Algebraic Generalisation from Concrete Cases: Learners have accumulated three or four concrete instances of rational + rational = rational (from the demonstrations and group work). The teacher now lifts this to an algebraic proof, making the pattern explicit and universal. This is a key NGSS Science and Engineering Practice	Collect the 'correction or confirmation' statements by asking 4–5 learners to read theirs aloud. Teacher listens for: (1) correct use of the term 'rational' in the corrected prediction; (2) reference to the p/q definition; (3) mention of the closure property (even if not by name). Written statements in books serve as a portfolio artefact

📖 **HTML:** [Areas of Combined Figures Involving Circles (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	stationery scenario. They then revisit the predictions on the board and individually write a 'correction or confirmation' statement: 'My prediction was ____ because ____ ; now I know ____.' To close the phase, the teacher poses one extension question: 'What about subtraction — will a − b always be rational if a and b are rational? Convince your partner in 45 seconds.'	sums of integers are integers. Is qn an integer and non-zero? YES. So a + b is ALWAYS rational.' Ask the class in a choral response: 'What do we call this property — when an operation on numbers in a set always stays in that set?' [WAIT TIME: 8 seconds.] Accept 'closure' if known; introduce the term if not. Say: 'Write this in your books with your own example from your group's scenario — I give you 3 minutes.' Cold-call 2 learners to share their written statements: 'Otieno, read your statement — tell us your example.' After the 3 minutes, ask the extension question about subtraction. [WAIT TIME: 45 seconds of partner talk.] Cold-call 1 pair: 'Halima and Mwenda — what did you agree on?' Confirm: subtraction of rationals is also always rational, using the same algebraic argument with a − b = (pn − mq)/(qn). Explicitly connect back to phenomenon: 'This is why Sister Wanjiku's average temperature calculation always gives her a neat, rational answer — she is adding and dividing rational numbers, and the result must stay rational.'	equivalent — using mathematics and computational thinking to construct an explanation. Writing the 'correction or confirmation' statement forces metacognitive reflection on the prediction made at the lesson's start.	for Lesson 4. Misconception watch: learners who write 'the answer is rational because the decimal ends' are partially correct but have not yet grasped the algebraic closure argument — note these learners for follow-up in Lesson 5.
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Worked example: Order of operations (PEMDAS)] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos	The class returns to the Driving Question Board displayed at the front. Each group writes one sticky note (or one item on their section of the board) responding to the new DQB question: 'Do combined operations on rationals always give a rational result?' Groups must write: (a) what they now believe the answer is, (b) one piece of evidence from today's lesson supporting that belief, and (c) one question they still have (e.g., 'What about multiplication?	Say: 'It is time to update our Driving Question Board. Today's focus question is written at the top: Do combined operations on rationals always give a rational result? Each group has 2 minutes to write your three-part sticky note — answer, evidence, new question.' [Set a 2-minute timer.] While groups write, circulate and prompt groups who are stuck: 'What did your scenario show you today?' After notes are posted, say: 'Take 30 seconds to walk up	Evidence-Claim-Question DQB Protocol: Each group must link their claim ('yes, always rational') to a specific evidence point from today's lesson before posting. This prevents the DQB from becoming a list of unsupported opinions and models the scientific practice of evidence-based reasoning. The class vote on 'still wondering' questions makes the DQB a living document that drives the lesson sequence forward rather than a static display.	Teacher photographs the DQB after the lesson. Assessment indicators: (1) all groups can articulate a claim with evidence — if any group only states a claim, teacher records this as a gap; (2) the quality of 'still wondering' questions reveals depth of thinking — surface questions ('what is BODMAS?') vs. generative questions ('does this work for subtraction of irrational numbers?'). The two voted questions inform teacher planning

📖 **HTML:** [Areas of Combined Figures Involving Circles (Flexbook)] Source: CK-12 🔍 [Search ARES for similar readings]	What about irrational numbers — can you add them to get rational?'). The class reads all sticky notes silently, then votes on the two most interesting 'still wondering' questions using a show of hands.	and read two other groups' notes silently.' [Allow movement.] Bring class back: 'I am going to read aloud the three most interesting new questions posted. Then we will vote on the two we most want to answer in future lessons.' Read the questions aloud. Say: 'Raise your hand for the question you most want us to investigate.' Tally votes and circle the top two questions on the board. Say: 'These two questions will travel with us into Lessons 5 and 6 — multiplication and division of rationals, and a first look at what happens when we mix rationals and irrationals.'		for Lessons 5–6.
Model Building Phase 📹 **VIDEO:** [Worked example: Order of operations (PEMDAS)] Source: Khan Academy (English - US curriculum) 🔍 [Search ARES for similar videos] 📖 **HTML:** [Areas of Combined Figures Involving Circles (Flexbook)] Source: CK-12 🔍 [Search ARES for similar readings]	Learners open their cumulative real-number diagram (begun in Lesson 1 and updated in Lessons 2 and 3). They add a new annotation layer labelled 'Operations on ℚ — Lesson 4'. In the rational numbers region of their diagram, they draw a small loop arrow (representing closure) and write next to it: 'ℚ + ℚ → ℚ' and 'ℚ − ℚ → ℚ (closure under + and −)'. Below the diagram, learners write one sentence connecting today's learning to the anchoring phenomenon: 'Sister Wanjiku's average temperature is rational because ___.'	Say: 'Take out your real-number diagram from Lesson 1. We are going to add one more layer today — do not erase anything you drew before, just add to it.' Project a sample diagram on the board (teacher's own version) and demonstrate adding the closure arrow and notation in real time: 'I draw a small loop inside the rational numbers bubble — this loop means: operations on rationals stay inside the rational numbers bubble. I write ℚ + ℚ → ℚ.' Give learners 3 minutes to update their diagrams. Say: 'Now write your one connection sentence at the bottom. Use the words rational, p/q, and terminates.' [WAIT TIME: allow 2 minutes for writing.] Cold-call 2 learners to read their sentences: 'Njeri, read your sentence — strong voice.' After both share, ask the class: 'Did anyone write something different from Njeri and Abubakar? Raise your hand.' Cold-call 1 different response. Close by saying: 'Our model is growing. In	Cumulative Model Revision: The real-number diagram is the lesson sequence's anchor artefact — each lesson adds a layer without erasing prior knowledge. The loop arrow for closure is a deliberate visual metaphor: operations on rationals circle back into the rational set. The connection sentence to the anchoring phenomenon forces learners to link the abstract algebraic property to the concrete health-centre data that opened the unit, maintaining storyline coherence across all eight lessons.	Teacher circulates during diagram annotation, checking: (1) the closure arrow is correctly placed inside (not outside) the rational numbers region; (2) the notation 'ℚ + ℚ → ℚ' is written correctly; (3) the connection sentence contains at least two of the three required terms (rational, p/q, terminates). Collect 4–5 exercise books at the end of the lesson for a quick portfolio check — look especially at the connection sentence for evidence that learners are linking today's algebra to the phenomenon rather than treating it as an isolated skill.

		Lesson 5 we will ask: does the closure loop hold for multiplication and division too? Keep this diagram safe — you will need it next lesson.'		

D. TEACHER REFLECTION
1. During the Observe phase group work, which groups struggled most with converting between fraction and decimal representations of the same rational number — and what does this tell me about gaps from Lesson 3 that I need to address in the warm-up of Lesson 5?\n2. Did my algebraic generalisation (a + b = (pn + mq)/(qn)) land with most learners, or did I move from concrete examples to abstraction too quickly? What evidence from the 'correction or confirmation' statements supports my answer?\n3. Were the three group case scenarios pitched at the right level of challenge — did any group finish too quickly (suggesting the numbers were too simple) or not reach a solution (suggesting the fractions were too complex)? How will I differentiate in Lesson 5?\n4. Looking at the DQB 'still wondering' questions that received the most votes, are learners' curiosities pointing toward the topics planned for Lessons 5–6 (multiplication/division of rationals), or are they pointing somewhere unexpected — and if unexpected, how do I honour that curiosity within the sub-strand sequence?\n5. How equitably did I distribute cold-call opportunities across gender, seating position, and confidence level today — and which learners have not yet spoken in front of the class across Lessons 1–4 and need to be deliberately included tomorrow?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	We observed that when Mama Akinyi's market prices (Ksh 45.50 + Ksh 12.75 + Ksh 38.00) and the Kisumu school stationery quantities were added and subtracted, every result was a number that could still be written as a fraction p/q — the answers always 'ended neatly' or repeated, never producing a non-terminating, non-repeating decimal like √2.
What did I learn?	We learned that rational numbers are closed under addition and subtraction: if a = p/q and b = m/n are both rational, then a + b = (pn + mq)/(qn) is also rational. BODMAS governs the order of combined operations. The closure property means that no matter how many rational numbers Sister Wanjiku adds or subtracts, her result will always remain rational.
How does this explain the phenomenon?	This helps explain the anchoring phenomenon: Sister Wanjiku's average temperature (36.8 + 37.0 + 37.5 ÷ 3) involves only addition and division of rational numbers. Because rational numbers are closed under these operations, the result must be rational — it terminates or repeats. The tile diagonal (√2), by contrast, is irrational and was never produced by adding or subtracting rational numbers; it came from a square root operation on a non-perfect square, which takes us outside the rational number set entirely.

LESSON 5 (40 minutes): Combined Operations on Rational Numbers II — Multiplication & Division	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Learners extend their understanding of combined operations on rational numbers to include multiplication and division, deepening their ability to distinguish between numbers that 'end neatly' (rational) and those that do not (irrational), and applying this understanding to authentic Kenyan agricultural contexts.
Knowledge	Learners know that multiplication and division of rational numbers — including mixed numbers and fractions — always produce rational results; they can convert mixed numbers to improper fractions before operating, and they can express results in simplest form.
Skills	Learners perform combined operations (multiplication and division) on rational numbers presented as mixed numbers and fractions, including multi-step problems drawn from Kenyan farming contexts; they annotate and critique worked examples for correctness and efficiency.
Attitudes	Learners appreciate that rational-number arithmetic is a precise, reliable tool for fair sharing and resource planning — skills of direct value to Kenyan farmers, co-operative members, and health workers; they demonstrate respect by taking turns and listening during gallery-walk annotations.
Key Inquiry Question	Why are numbers important?
Purpose in Storyline	In Lessons 1–4, learners classified real numbers and performed addition and subtraction of rational numbers. Lesson 5 extends those operations to multiplication and division, using Kenyan agricultural scenarios (Kericho tea and Rift Valley maize) to show that rational × rational and rational ÷ rational always land on a number that 'ends neatly' — advancing the driving question by reinforcing the boundary between rational and irrational numbers through new operations.
Safety Notes	No physical hazards. Remind learners to handle printed gallery-walk posters with care and to use markers only on designated annotation strips, not on walls.

B. LESSON OVERVIEW

In this 40-minute Observe + Explain lesson, learners explore multiplication and division of rational numbers (including mixed numbers) through two Kenyan agricultural scenarios: a Kericho tea farmer dividing $3\frac{3}{4}$ kg of dried tea leaves equally among 5 co-operative members, and a Rift Valley maize farmer multiplying a seed rate of $2\frac{2}{5}$ kg/acre over $3\frac{1}{2}$ acres. The core activity is a structured Gallery Walk: four fully worked problems are posted around the room; learner pairs rotate every 5 minutes, reading, verifying, and annotating each solution with corrections or alternative methods on sticky-note strips. The teacher then cold-calls 3 pairs to defend their annotations before the class consolidates a shared method. The Driving Question Board is updated to record the new evidence that multiplying or dividing rational numbers always yields a rational result — another clue in explaining why the health volunteer's temperature average 'ends neatly' while $\sqrt{2}$ does not.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase	Learners read two short scenario cards displayed on the board	Display the two scenario cards. Say: 'Before we calculate anything,	Prediction Journaling with Phenomenon Anchor: by explicitly	Scan written predictions for two observable indicators: (1) learner

📹 **VIDEO:** [Identifying Mixed Numbers and Improper Fractions - Example 1] Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** [Properties of Rational Numbers (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	(Kericho tea farmer: 3¾ kg ÷ 5 members; Rift Valley maize farmer: 2⅖ kg/acre × 3½ acres). Working individually for 2 minutes, they record their predicted answers and predict whether each answer will 'end neatly' or 'go on forever', writing their reasoning in one sentence in their exercise books. Pairs then whisper-share predictions for 1 minute.	I want you to predict — will these answers end neatly, like the health volunteer's average temperature, or go on forever like √2? Write your prediction AND your reason. You have 2 minutes — go.' Circulate silently for 2 minutes observing written predictions; do not confirm or deny. After whisper-share, cold-call 2 learners (one per scenario): 'Tell us your prediction and your one-sentence reason.' After each, say: 'Hold that thought — the gallery walk will tell us if you are right.'	linking each scenario to the driving-question tension ('ends neatly vs. goes on forever') before any calculation, learners activate prior knowledge from Lessons 1–4 and commit to a testable claim, creating cognitive dissonance that motivates careful observation during the gallery walk.	writes a specific numerical estimate (e.g., 'about ¾ kg') rather than just 'yes/no'; (2) learner's reasoning references fractions or 'rational' from prior lessons. Flag pairs who predict irrational results for targeted follow-up during the gallery walk.
Observe Phase 📹 **VIDEO:** [Identifying Mixed Numbers and Improper Fractions - Example 1] Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** [Properties of Rational Numbers (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Four gallery-walk stations are set up around the room, each with a large A3 poster showing a fully worked problem and a blank annotation strip (sticky notes or a ruled strip of paper). Pairs rotate to each station every 5 minutes (teacher signals with a clap). At each station they: (a) read the worked solution carefully, (b) check every step for correctness, (c) write on the annotation strip either a correction, an alternative method, or a 'confirmed correct ✓ because…' note with their initials. Station problems: Station 1 — Kericho tea: 3¾ ÷ 5 (division of mixed number by whole number). Station 2 — Rift Valley maize: 2⅖ × 3½ (multiplication of two mixed numbers). Station 3 — A Kisumu fish trader buys 4½ crates and sells ⅔ of the stock; how many crates sold? (multiplication with fraction of a mixed number). Station 4 — A Mombasa coconut seller shares 6⅗ kg of copra equally among 4 buyers; what does each receive? (division of mixed number).	Before rotation begins, say: 'Your job at each station is to be a quality-control inspector — like a KEBS officer checking a product. Read every step. If you find an error, write the correct step and explain why. If the solution is correct, write why it is correct. You are not just copying — you are evaluating.' Set a visible 5-minute timer. Signal each rotation with a single clap and say: 'Rotate — move to the next station clockwise.' During rotations, circulate and observe annotations; do not correct learners directly. Whisper to pairs who are only writing 'correct ✓' without reasoning: 'Tell me why it is correct — what rule did they use?' After all 4 rotations, say: 'Return to your seats. You have 1 minute to re-read your own annotations silently.' Allow 1 minute of silent review.	Structured Gallery Walk with Annotation: this strategy shifts the epistemic authority from the teacher to the mathematical work itself. Learners must engage with reasoning at the step level (not just the answer), mirroring how scientists evaluate evidence — connecting to NGSS Science and Engineering Practice 7 (Engaging in Argument from Evidence). The four Kenyan contexts (tea, maize, fish, coconut) reinforce that rational ÷ rational and rational × rational always produce rational results — observable evidence for the driving question.	Collect and scan annotation strips after the activity. Observable indicators of understanding: (1) learner correctly identifies and fixes an error in a multi-step solution (e.g., failing to convert mixed number to improper fraction before multiplying); (2) learner's 'confirmed correct' note names the specific rule applied (e.g., 'multiply numerators then denominators'); (3) learner's annotation uses the word 'rational' or 'fraction' at least once. Indicators of misconception: learner 'corrects' a step that was actually right, or skips the invert-and-multiply step in division.

Explain Phase 📹 **VIDEO:** Identifying Mixed Numbers and Improper Fractions - Example 1 Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Properties of Rational Numbers (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	The class reconvenes. The teacher selects 3 gallery-walk annotations (one correct confirmation, one valid correction, one alternative method) and projects or reads them aloud. The 3 pairs who wrote those annotations are cold-called to defend their reasoning to the class. After each defence, the class votes silently (thumbs up / thumbs sideways / thumbs down) on whether they agree. The teacher then facilitates a whole-class consolidation: on the board, works through the standard method for multiplying and dividing mixed numbers step-by-step, co-constructing the steps with learner input. Learners copy the consolidated method into their exercise books and then individually solve one exit problem: 'A Nakuru dairy farmer has 5⅓ litres of milk and fills bottles of ⅔ litre each. How many full bottles does she fill?' (5⅓ ÷ ⅔).	Cold-call Pair A (correct confirmation): 'Pair A, read your annotation for Station 1 aloud and defend it — why is that step correct?' Allow 15 seconds WAIT TIME before prompting. After response, say: 'Class, thumbs up if you agree, sideways if unsure, down if you disagree. [Pause 5 seconds.] [Name a 'sideways' learner] — tell us your doubt.' Repeat for Pair B (correction) and Pair C (alternative method). During consolidation, write each step on the board and ask: 'What do I do FIRST when I see a mixed number in a multiplication or division problem?' WAIT 10 seconds. Cold-call a learner who has not yet spoken. Build steps: (1) Convert to improper fractions. (2) For division, invert the divisor and multiply. (3) Multiply numerators; multiply denominators. (4) Simplify. After learners copy, say: 'Now solve the Nakuru dairy problem alone — 3 minutes. Show every step.' Circulate and note common errors.	Claim–Evidence–Reasoning (CER) Defence: each cold-called pair must state a claim (this step is correct/incorrect), cite evidence (the numbers in the worked solution), and give reasoning (the mathematical rule). This mirrors NGSS Practice 6 (Constructing Explanations) and makes the sensemaking process public and critiqueable. The consolidation then formalises the pattern, connecting it back to the phenomenon: 'Every answer we got today was a fraction or a terminating decimal — a rational number. None of them went on forever like √2.'	Exit problem (Nakuru dairy farmer): collect exercise books or observe work in real time. Indicators of mastery: (1) learner correctly converts 5⅓ to 16/3 and ⅔ remains as is; (2) learner inverts ⅔ to 3/2 and multiplies; (3) arrives at 8 full bottles with correct simplification. Common error to watch: learner divides numerator by numerator and denominator by denominator (12÷2=6, 3÷3=1 → wrong answer of 6). Flag these learners for follow-up in Lesson 6.
Driving Question Board (DQB) Creation 📹 **VIDEO:** Identifying Mixed Numbers and Improper Fractions - Example 1 Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Properties of Rational Numbers (Flexbook) Source: CK-12	Learners return to the class Driving Question Board. Each pair receives one sticky note. They write one new piece of evidence their lesson has given them about the driving question: 'Why do some numbers end neatly while others go on forever — and how does knowing the difference help us make better decisions?' Pairs post their sticky notes under the column 'What we now know'. Two learners are invited to read their notes aloud. The teacher highlights the emerging pattern across Lessons 1–5 by pointing to the board: addition, subtraction, multiplication, and division of rational numbers always produce	Hand out sticky notes. Say: 'In one or two sentences, write the most important thing today's lesson adds to our answer to the driving question. Mention the word rational or irrational.' Allow 2 minutes of writing. After posting, read two contrasting notes aloud and say: 'Notice — both pairs are saying that our Kericho and Rift Valley results were rational. Does that surprise anyone, given your prediction at the start?' WAIT 10 seconds. Cold-call one learner. Then gesture to the full DQB: 'Look at what we have built over five lessons. Every operation we have tested on rational numbers gives us a rational result. So why	DQB Cumulative Mapping: the DQB functions as a public, growing evidence wall — a direct analogue to a scientific investigation log. By explicitly mapping today's evidence (multiplication and division of rationals are rational) onto the board alongside prior lessons' evidence (addition and subtraction of rationals are rational), learners build a coherent conceptual structure rather than isolated facts. The unresolved tension about √2 sustains motivation for subsequent lessons.	Read all posted sticky notes after class. Indicators of conceptual progress: (1) learner explicitly states that the result of multiplying or dividing rational numbers is rational; (2) learner connects this back to the health volunteer's temperature data or another phenomenon example; (3) learner identifies the remaining open question (why √2 is different). Notes that only describe procedure ('we learned to multiply fractions') indicate surface-level learning and signal need for deeper conceptual questioning in Lesson 6.

🔍 Search ARES for similar readings	rational numbers — the health volunteer's averages always end neatly because they are combinations of rational measurements.	does √2 still go on forever? That tension is still alive — and we will keep investigating.'		
Model Building Phase 📹 **VIDEO:** Identifying Mixed Numbers and Improper Fractions - Example 1 Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Properties of Rational Numbers (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners open their running 'Real Numbers Model' — a concept diagram started in Lesson 1 (could be a hand-drawn number-line-based map or a table in their exercise books). They add a new branch or row: 'Operations on Rational Numbers — Multiplication & Division', recording: (a) the rule in words ('rational × rational = rational; rational ÷ rational = rational, provided divisor ≠ 0'), (b) one worked example from today's gallery walk (their choice), and (c) a note on how this connects to the phenomenon ('the health volunteer's average temperature is rational because she adds and divides rational measurements — her result always ends neatly'). Pairs compare their updated models and identify any difference in how they drew the connection to the phenomenon.	Say: 'Open your Real Numbers Model from Lesson 1. You are going to add today's evidence. Three things to include — the rule, one example, and the connection to our health volunteer at Mathare North. You have 3 minutes.' Circulate and look for models that omit the phenomenon connection. Prompt those pairs: 'How does what you learned today explain why the health worker's average temperature was not like √2?' After 3 minutes, say: 'Show your model to your partner — does their model say the same thing in a different way? Point out one thing you like about their model.' Allow 1 minute of pair comparison. Close by saying: 'Your model is growing. In Lesson 6, we will investigate reciprocals — and ask whether the reciprocal of a rational number is also rational. Keep that question in mind overnight.'	Cumulative Concept-Map Revision (NGSS Practice 2 — Developing and Using Models): requiring learners to physically update a model they have maintained since Lesson 1 makes conceptual growth visible and personal. The three-part addition (rule + example + phenomenon link) prevents the model from becoming a mere formula list and keeps the anchoring phenomenon alive as the organising thread of the sub-strand.	Circulate and photograph or note the model updates of 4–5 pairs. Indicators of strong model building: (1) rule is stated in words, not just as a worked example; (2) phenomenon connection is explicit and specific (mentions 'health volunteer', 'temperature', or 'Mathare North'); (3) model visually distinguishes between rational results (today) and the irrational √2 introduced in Lesson 1. Weak indicator: model only contains a copied worked example with no rule or phenomenon link — these learners need a structured model-building scaffold in Lesson 6.

D. TEACHER REFLECTION

After the lesson, reflect on the following: (1) Gallery Walk quality — did at least 70% of annotation strips contain a named rule or a genuine correction, or did most pairs only write 'correct ✓'? If annotations were superficial, consider adding a structured annotation template (Claim / Step number / Rule used / My verdict) for Lesson 6. (2) Cold-call depth — did the 3 cold-called pairs articulate reasoning beyond 'because the answer is right'? If not, rehearse the CER (Claim–Evidence–Reasoning) sentence frame with the class before the next gallery walk. (3) Exit problem results — how many learners correctly solved the Nakuru dairy problem in 3 minutes? If fewer than 60% reached the correct answer, open Lesson 6 with a 5-minute re-teach of the invert-and-multiply step using a visual (drawing $5\frac{1}{3}$ litres as a number line partitioned into $\frac{2}{3}$-litre segments). (4) DQB connections — did learner sticky notes link today's results to the driving question phenomenon, or did they remain procedural? The driving question thread must be explicitly re-stated at the start of every lesson if learners are not spontaneously connecting to it. (5) Equity check — note which learners have not yet been cold-called across Lessons 1–5; ensure full rotation by Lesson 8.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners observed four fully worked multiplication and division problems in a gallery walk — a Kericho tea farmer dividing 3¾ kg among 5 co-operative members, a Rift Valley maize farmer multiplying 2⅖ kg/acre over 3½ acres, a Kisumu fish trader finding ⅔ of 4½ crates, and a Mombasa coconut seller sharing 6⅗ kg among 4 buyers — and annotated each solution for correctness and alternative methods.
What did I learn?	Learners learned that multiplying or dividing rational numbers (including mixed numbers) always produces a rational number; that the correct procedure requires converting mixed numbers to improper fractions first, and inverting the divisor before multiplying in division problems; and that every agricultural result in today's lesson 'ended neatly', just like the health volunteer's average temperature.
How does this explain the phenomenon?	Learners explained, through gallery-walk annotations and cold-call defences using Claim–Evidence–Reasoning, why each step in a worked solution was correct or incorrect; they consolidated the standard method for multiplying and dividing mixed numbers; and they updated their running Real Numbers Model to record that all four arithmetic operations on rational numbers yield rational results, deepening the distinction between rational and irrational numbers anchored in the driving question.

LESSON 6 (40 minutes): Reciprocals of Real Numbers — What Happens When You "Undo" a Number on a Calculator?	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Learners use calculators to find reciprocals of integers, fractions, and decimals — and confront the question of whether the reciprocal of an irrational number is itself irrational — advancing their understanding of how every real number in the health volunteer's data set relates to its multiplicative inverse.
Knowledge	– The reciprocal of a non-zero real number x is 1/x, and the product x × (1/x) = 1 always. – The reciprocal of a rational number (p/q, q ≠ 0) is q/p, which is also rational. – The reciprocal of an irrational number is irrational because if 1/x were rational (say r/s), then x = s/r would also be rational — a contradiction.
Skills	– Use a calculator correctly and responsibly to find reciprocals of integers, fractions, and decimals drawn from the Mathare North health-centre data. – Identify and record the pattern x × (1/x) = 1 from a systematic table of results. – Construct a logical argument (proof by contradiction) for why the reciprocal of an irrational number must itself be irrational.
Attitudes	– Demonstrate responsibility in the careful and accurate use of a calculator as a mathematical tool. – Appreciate that mathematical relationships such as x × (1/x) = 1 hold universally across all categories of real numbers.
Key Inquiry Question	Why are numbers important?
Purpose in Storyline	Lessons 4 and 5 showed that adding, subtracting, multiplying, and dividing rational numbers always produces a rational number. Lesson 6 now asks: what happens when we 'undo' a number — find its reciprocal? Returning to the blood pressure readings (120/80, 135/90) and the tile diagonal (√2) from the anchoring phenomenon, learners discover that the reciprocal operation preserves rationality for rational numbers and preserves irrationality for irrational ones, deepening the boundary between the two sets.
Safety Notes	No physical hazards. Remind learners to handle shared calculators with care; avoid pressing keys forcefully. If using shared school devices, sanitise hands before and after use.

B. LESSON OVERVIEW

In this sixth lesson of the sequence, learners return to the blood pressure readings that the Mathare North Health Centre volunteer recorded — 120/80 and 135/90 — and ask a new question: what is the 'opposite' of each number in the multiplicative sense? Building on their mastery of multiplication and division of rational numbers from Lesson 5, learners now formalise the idea of the reciprocal (multiplicative inverse) as the number that, when multiplied by the original, gives exactly 1. Using calculators, they systematically investigate reciprocals of integers (e.g., 80, 90, 3), unit fractions (e.g., 1/37, 2/5), and the decimals that appeared in the health-centre data (36.8, 37.5). Every result is recorded in a structured table, and learners confirm the universal pattern x × (1/x) = 1.

The lesson's intellectual climax arrives when learners ask: "What about √2 — the irrational tile diagonal from our anchoring phenomenon? Does it have a reciprocal, and is that reciprocal also irrational?" Through guided reasoning (an accessible proof by contradiction), learners convince themselves that if 1/√2 were rational, then √2 itself would have to be rational — but we proved in Lesson 3 that it is not. Therefore 1/√2 must be irrational. This insight enriches the class model of the real number system: the reciprocal operation keeps rational numbers rational and keeps irrational numbers irrational.

Throughout the lesson, the KICD value of Responsibility is made explicit: responsible calculator use — reading the display carefully, rounding only when instructed, and not confusing a truncated screen display for a terminating decimal — is framed as a mathematical integrity skill, not merely a care-for-equipment rule. Learners update their Driving Question Board and revise their cumulative model of real numbers to include reciprocals as a new layer of structure.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** Finding the Reciprocal of Fractions Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Identification and Writing of Reciprocal Fractions (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Each learner receives a half-page 'Predict Card' listing five numbers drawn directly from the anchoring phenomenon and previous lessons: 80 (the denominator of the blood pressure reading 120/80), 2/5, 36.8 (a patient temperature), 0 (teacher's deliberate trap), and √2 (the tile diagonal). Without a calculator, learners write their best prediction for 'the number I would multiply each of these by to get exactly 1' — or write 'impossible' if they believe no such number exists. They also predict whether that reciprocal will be rational or irrational, circling R or I next to each answer. Learners share predictions with a shoulder partner for 90 seconds before the class discussion.	Distribute Predict Cards face-down. Say: 'Before we touch a calculator today, I want your mathematical instincts. Flip your card over — you have 3 minutes to write your predictions. No calculator yet — use what you already know.' [3-minute silent individual work.] After individual work, say: 'Turn to your neighbour — compare your prediction for √2. Do you both agree on whether its reciprocal is rational or irrational? You have 90 seconds.' [Circulate; listen for misconceptions — many learners will predict 1/√2 is rational.] After partner talk, cold-call 4 learners: first two on the number 80, next two on √2. Ask exactly: 'What number did you predict for 80, and how did you reason?' [WAIT TIME: 12 seconds after each nomination before accepting an answer.] For the zero entry, ask: 'Did anyone write anything other than impossible for zero?' [WAIT TIME: 10 seconds.] Record class predictions visibly on the board in a two-column table (Number | Predicted Reciprocal). Tell learners: 'We are going to test every prediction with a calculator in the Observe phase — and one of these will surprise most of you.'	Prediction-Before-Evidence (Anticipatory Thinking): By committing predictions to paper before gathering data, learners activate prior knowledge of multiplication and division, surface misconceptions about zero and irrational numbers, and create a cognitive 'tension' that makes the calculator results more memorable. The shoulder-partner share ensures every learner has articulated a position before class discussion.	Teacher looks for: (1) Whether learners correctly predict reciprocals of integers and simple fractions (indicates readiness for the pattern work). (2) Whether learners write 'impossible' for zero — if not, this is a critical misconception to address explicitly in the Explain phase. (3) How learners reason about √2 — any learner who says 'I think 1/√2 is rational' has a productive misconception that will drive the lesson's key discussion. Teacher notes 2–3 specific learner predictions on the board to revisit at the end of the Observe phase.
Observe Phase	Working in pairs, learners receive a printed Reciprocal Investigation	Distribute calculators and tables simultaneously. Say: 'In this	Structured Data-Recording Table with Pattern Identification: The	Teacher circulates and checks: (1) Are learners recording all digits

📹 **VIDEO:** [Finding the Reciprocal of Fractions] Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** [Identification and Writing of Reciprocal Fractions (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Table with three sections. Section A lists integers: 80, 90, 3, 120, 37. Section B lists fractions and decimals: 2/5, 3/8, 36.8, 37.5, 120/80 (treated as the fraction 3/2). Section C lists irrational-adjacent entries: √2 (enter as 2, then √, then use 1÷ key), √3, and π (use the π key). For each number x, learners calculate 1÷x on the calculator, record all displayed digits in column 2, then calculate x × (their recorded reciprocal) in column 3, and record the result. They also fill in column 4: 'Is the reciprocal terminating, repeating, or non-terminating non-repeating?' — determined by inspecting the display and cross-referencing with their knowledge from Lessons 2–3. A guiding question at the bottom of the table reads: 'What do you notice about the number in column 3 every single time?'	phase, the calculator is your measuring instrument — treat it the way Sister Wanjiku at Mathare North treats her thermometer: read it carefully, record every digit you see, and do not round unless I specifically ask you to.' Demonstrate the key sequence for one example (x = 3) on the board using a projected calculator image: press 1 ÷ 3 =, read '0.333333333', enter it in column 2, then multiply 3 × 0.333333333 = and note the result is '1' (or '0.999999999' on some calculators — address this immediately). Say: 'If your calculator shows 0.999999999 instead of 1, that is a rounding artefact — mathematically the product IS exactly 1. This is exactly why responsible calculator use matters.' [Circulate for 8 minutes of pair work.] At the 5-minute mark, call a 30-second pause: 'Pairs, hold your work. Cold question — [name Learner A]: What did you get when you multiplied 36.8 by its reciprocal?' [WAIT TIME: 12 seconds.] Then: '[Name Learner B]: What about for √2 — what does column 4 say?' [WAIT TIME: 12 seconds.] After all sections are complete (approx. 10 minutes total), ask all learners to underline column 3. Say: 'What is true about every single entry in column 3?' Accept answers — confirm the universal result: 'x × (1/x) = 1 for every non-zero real number, rational or irrational.' Write this prominently on the board.	three-section table scaffolds learners from familiar integers (Section A) to decimals (Section B) to irrational numbers (Section C), so the pattern x × (1/x) = 1 is established with high confidence before it is tested on the harder irrational case. Column 3 makes the pattern visible as a column of 1s, creating a powerful visual anchor. The deliberate inclusion of the blood pressure fractions (120/80 = 3/2) connects the activity back to the anchoring phenomenon.	from the calculator display, or are they rounding prematurely? (Premature rounding is a responsible-use failure — address it immediately.) (2) Do column 3 entries consistently show 1? If a pair gets wildly different values, investigate key-press errors. (3) In Section C, does the learner correctly identify √2 and π reciprocals as non-terminating and non-repeating in column 4? This is the bridge to the Explain phase discussion about irrationality of reciprocals. Target at least 4 pairs for a 30-second check-in during circulation.
Explain Phase 📹 **VIDEO:** [Finding the Reciprocal of Fractions]	Learners engage in two consecutive sensemaking activities. First, they complete a 'Pattern Statement' in their exercise books: using the table	Begin by saying: 'Look at column 3 in your table. I want someone to give me a sentence — not just a number — that describes what always happened.' Cold-call 2	Proof by Contradiction as Accessible Logic: Rather than presenting the irrationality of 1/√2 as a fact to memorise, the teacher walks learners through the	Teacher checks written Pattern Statements — look for: (1) Does the statement explicitly name 'multiplicative inverse' or 'reciprocal'? (2) Is the condition

Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Identification and Writing of Reciprocal Fractions (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	data as evidence, they write a generalisation in their own words, then compare it to the formal statement the teacher writes on the board: 'The reciprocal of a non-zero real number x is 1/x, and x × (1/x) = 1.' They annotate each part. Second, learners work through a teacher-guided logical argument about irrationality of reciprocals: the teacher poses the question 'Could 1/√2 be rational?' and walks learners through a proof by contradiction — if 1/√2 = p/q (rational), then √2 = q/p (also rational) — but we proved in Lesson 3 that √2 is irrational. Contradiction! Therefore 1/√2 must be irrational. Learners write this argument in their books in their own words. Finally, learners revisit the health-centre context: they calculate the reciprocal of the average temperature 37.1 °C (computed as (36.8 + 37.0 + 37.5) ÷ 3 — which they verify equals 111.3/3 = 37.1, a terminating rational), confirm that 1/37.1 = 10/371 (rational), and contrast this with 1/√2 (irrational).	learners. [WAIT TIME: 12 seconds each.] Write the best student-generated sentence on the board, then refine it to the formal definition. Say: 'This relationship — x times its reciprocal equals 1 — is called the multiplicative inverse property. Write it as a heading in your notes.' Then pivot: 'Now — the big question. We have been asking all lesson: is the reciprocal of an irrational number also irrational? Let me show you how mathematicians decide this.' Write on the board: 'Suppose 1/√2 = p/q where p and q are integers and q ≠ 0.' Ask: 'What does that mean about √2?' [WAIT TIME: 15 seconds — allow thinking time.] Guide learners to rearrange to √2 = q/p. Then ask: 'But what did we prove in Lesson 3 about √2?' [Cold-call 1 learner.] Confirm: 'We proved √2 is irrational — it CANNOT equal q/p. So our assumption was wrong. 1/√2 cannot be rational. It must be irrational.' Explicitly address the Responsibility value: 'Now — why does this matter for responsible calculator use? Your calculator showed 1/√2 ≈ 0.70710678. That display ENDS — but is it actually terminating?' [WAIT TIME: 12 seconds.] Confirm: 'No. The calculator truncated an infinite non-repeating decimal. A responsible calculator user knows the display is an approximation, not the exact value.' Conclude by computing 1/37.1 on the projector: 'Sister Wanjiku's average temperature was 37.1 °C. Its reciprocal is approximately 0.02695. Is that rational?' [Cold-call 1 learner, WAIT TIME 12 seconds.] Confirm: 37.1 = 371/10, so its reciprocal is 10/371 —	contradiction structure ('suppose the opposite is true… but that contradicts something we already know… therefore our supposition was false'). This is learners' first encounter with this reasoning style in the sequence; framing it as 'how mathematicians decide' empowers learners to see mathematical argument as a tool they can use. Connecting the proof back to Lesson 3's result reinforces storyline coherence.	'non-zero' included? (3) In the proof-by-contradiction notes, can the learner state WHY the contradiction arises — i.e., they must reference the Lesson 3 result that √2 is irrational? Ask 2–3 learners to read their contradiction arguments aloud; listen for whether they can articulate the logical chain without just copying board notes word-for-word.

		rational.		
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Finding the Reciprocal of Fractions] Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** [Identification and Writing of Reciprocal Fractions (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	Learners take out their DQB sticky notes (or the class DQB is projected). Each learner writes one 'We now know' card (green) and one 'We still wonder' card (yellow). The 'We now know' card must reference reciprocals and connect to either the blood pressure data or the √2 tile diagonal. Example green card: 'We now know that 1/(120/80) = 80/120 = 2/3 — a rational number — so blood pressure ratios have rational reciprocals.' Example yellow card: 'We still wonder — is there any operation on irrational numbers that produces a rational number?' Learners share cards with a partner before posting. The teacher then reads two or three posted cards aloud and uses them to narrate progress on the Driving Question.	Say: 'Take two minutes — write your green card and your yellow card. Your green card MUST mention either a blood pressure number or √2. Your yellow card should push our investigation forward.' [2 minutes silent writing.] After writing, say: 'Before you post, tell your partner what you wrote and make sure your green card actually answers part of our driving question.' [90 seconds partner share.] Invite learners to post cards on the class DQB. Read two green cards and one yellow card aloud. Point to the DQB and say: 'Look at how far we have come. In Lesson 1 we didn't know whether Sister Wanjiku's numbers were rational or irrational. Now we know: her temperatures and blood pressure readings are rational, their reciprocals are rational, and the tile diagonal's reciprocal is irrational. But our yellow cards show we are not done yet.' Explicitly reference the Driving Question text visible in the classroom: 'Why do some numbers end neatly while others go on forever — and how does knowing the difference help us make better decisions?'	Claim-Evidence-Reasoning (CER) on the DQB: The green/yellow card structure enforces a mini-CER cycle — learners must state a claim ('We now know…'), implicitly cite evidence (from the calculator table or the proof), and connect to the phenomenon (blood pressure or tile). The teacher's oral narration of selected cards models how to synthesise individual observations into a coherent storyline arc.	Teacher reads all posted green cards and checks: (1) Does the card state a correct mathematical claim? (2) Does it connect to the phenomenon (health centre data or tile diagonal) rather than being an abstract statement? Yellow cards are scanned for questions that reveal lingering misconceptions (e.g., 'Is 1/0 an irrational number?' — a productive confusion to address briefly). Cards with errors are not corrected publicly but are noted by the teacher for follow-up in Lesson 7.
Model Building Phase 📹 **VIDEO:** [Finding the Reciprocal of Fractions] Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:**	Learners retrieve their cumulative Real Number System model — a diagram they have been building since Lesson 1 (a nested set diagram: Real Numbers ⊃ Rational ⊃ Integers ⊃ Whole Numbers, with Irrational Numbers as a separate region inside Real Numbers but outside Rational). Today they add a new annotation layer: reciprocal arrows. Using a coloured pen (different from	Say: 'Open your Real Number models from Lesson 1. Today we add a new layer — reciprocals. Use a different colour so the progression is visible.' Display a completed teacher model on the projector, but with the reciprocal annotations blank — learners must decide where and how to annotate. Say: 'Think about where to draw the reciprocal arrow for √2 — inside which region? How do	Cumulative Model Revision with Colour-Coding: The progressive annotation strategy (each lesson's additions in a new colour) creates a visible record of conceptual growth that learners can review across all eight lessons. The explicit instruction to use arrows and labels — rather than just adding text — requires learners to represent a relationship (the reciprocal operation as a mapping	Teacher circulates and checks model annotations for: (1) Are the reciprocal arrows placed inside the correct regions (rational ↔ rational; irrational ↔ irrational)? (2) Is zero correctly excluded with 'NO RECIPROCAL'? (3) Do the two Model Update Sentences accurately describe what is new — i.e., do they say something beyond what was in Lessons 4–5 models? Collect 4–5 models briefly at the

Identification and Writing of Reciprocal Fractions (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	previous lessons), learners draw a small double-headed arrow labelled '×(1/x)=1' within the Rational region, annotated with an example from the health-centre data (e.g., '37.1 ↔ 10/371'). They draw a second such arrow within the Irrational region, annotated '√2 ↔ 1/√2 (both irrational)'. They also add the label 'NO RECIPROCAL' with a cross symbol near the origin point of the diagram, representing zero. Learners write a two-sentence 'Model Update Note' below their diagram explaining what is new since Lesson 5.	you know?' [WAIT TIME: 15 seconds.] Cold-call 1 learner to explain their placement reasoning. After learners complete annotations (approximately 4 minutes), say: 'Write exactly two sentences below your diagram — sentence 1: what the reciprocal operation does to a real number; sentence 2: how this connects to Sister Wanjiku's health-centre data.' [2 minutes.] Cold-call 2 learners to read their sentences. Affirm strong connections to the phenomenon. Conclude: 'Our model is now richer — it shows not just what kinds of numbers exist, but what happens when we perform the reciprocal operation within each category. This is how mathematicians build understanding: layer by layer, always asking whether a new operation respects the boundaries between number types.'	within a set), which is a higher-order representational skill than listing facts.	end of class to check written sentences; return at the start of Lesson 7.

D. TEACHER REFLECTION

1. During the Observe phase, did learners correctly identify the reciprocals of decimals like 36.8 and 37.5 as rational (terminating), and were they able to articulate why — using the p/q definition from Lesson 3 — rather than just reading the calculator display? If not, what scaffolding would help in Lesson 7?

2. How well did learners follow the proof-by-contradiction argument for the irrationality of 1/√2? Were they able to write it back in their own words, or did most simply copy the board? If the argument was unclear, consider building a second 'suppose the opposite' example (e.g., suppose 1/π is rational) as a warm-up in Lesson 7.

3. Did the Responsibility value around calculator use feel integrated into the mathematics, or did it feel like a separate 'values lesson'? Specifically: did the discussion of the truncated calculator display for 1/√2 help learners understand that a finite display is not proof of a terminating decimal?

4. Were learners able to connect the reciprocal concept back to the blood pressure fractions (120/80 and 135/90) in their DQB green cards and model annotations, or did most learners use generic numbers? If the phenomenon connection was weak, plan a more explicit phenomenon-return prompt at the start of Lesson 7.

5. Look at the yellow 'We still wonder' cards posted on the DQB. Do they reveal any mathematical questions that should be addressed before the sequence closes in Lesson 8 — for example, questions about reciprocals of very large or very small numbers, or whether combined operations on irrational numbers can ever produce rationals? Use these to shape the Lesson 7 DQB discussion.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners used calculators to find the reciprocal of every number in the health-centre data set — body temperatures (36.8, 37.0, 37.5), blood pressure values (120, 80, 135, 90), and the irrational tile diagonal $\sqrt{2}$ — and recorded that in every case the product $x \times (1/x)$ equalled exactly 1.
What did I learn?	The reciprocal (multiplicative inverse) of any non-zero real number x is $1/x$, and $x \times (1/x) = 1$ always holds. The reciprocal of a rational number is rational (e.g., $1/37.1 = 10/371$). The reciprocal of an irrational number is also irrational — proved by contradiction: if $1/\sqrt{2}$ were rational, $\sqrt{2}$ would be rational, contradicting Lesson 3. Zero has no reciprocal. A calculator display that 'ends' does not prove a decimal terminates — responsible calculator use requires knowing the difference between a truncated display and a truly terminating value.
How does this explain the phenomenon?	Sister Wanjiku's health-centre numbers — temperatures and blood pressure readings — are all rational, so their reciprocals are also rational and 'end neatly'. The tile diagonal $\sqrt{2}$ is irrational, and its reciprocal $1/\sqrt{2}$ is also irrational — it never ends and never repeats, even though a calculator makes it look finite. This means the rational/irrational boundary is preserved under the reciprocal operation, just as Lessons 4 and 5 showed it is preserved under addition, subtraction, multiplication, and division.

LESSON 7 (40 minutes): Real Numbers in Real-Life Applications & DQB Consolidation	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Learners consolidate their understanding of rational and irrational numbers by applying them to authentic Kenyan real-life tasks — computing average body temperature at a community dispensary, costing githeri ingredients for a school canteen, and classifying a fence-post length as rational or irrational — and then revisit the class Driving Question Board to answer accumulated questions with evidence gathered across Lessons 1–6.
Knowledge	Learners know that: (a) rational numbers can be expressed as exact fractions and their decimals either terminate or repeat; (b) irrational numbers such as $\sqrt{2}$ and $\sqrt{5}$ produce non-terminating, non-repeating decimals; (c) combined operations on rational numbers follow BODMAS; (d) the real-number system comprises both rational and irrational numbers.
Skills	Learners are able to: (a) perform combined operations on rational numbers in context; (b) determine whether a given number is rational or irrational and justify the decision; (c) approximate irrational numbers to a specified number of significant figures using a calculator; (d) communicate mathematical reasoning clearly to peers.
Attitudes	Learners appreciate the use of real numbers in everyday Kenyan life — in health, nutrition, and construction — and develop responsibility by taking care of calculators and respecting peers' contributions during group presentations.
Key Inquiry Question	Why are numbers important?
Purpose in Storyline	In Lesson 7 learners shift from acquiring new concepts (Lessons 1–6) to applying and consolidating them. Three carefully chosen real-life tasks mirror the anchoring phenomenon: Task A echoes the health worker's average-temperature calculation; Task B grounds rational-number operations in the familiar school-canteen context of githeri ingredients; Task C revisits the irrational $\sqrt{2}$ fence-post puzzle, now with $\sqrt{5}$, demanding classification and approximation. Group presentations and whole-class DQB consolidation make the full rational-vs-irrational tension explicit and prepare learners for the final lesson.
Safety Notes	Calculators are shared equipment — learners handle them with care and return them undamaged. No hazardous materials are used. Emotional sensitivity is observed when any personal measurement data (weight, height) arises in discussion.

B. LESSON OVERVIEW

Lesson 7 of 8 is a double-function lesson: an evidence-gathering application session and a Driving Question Board (DQB) Consolidation. Learners work in groups of 4–5 on three real-life application tasks that draw on all concepts covered in Lessons 1–6 (classifying real numbers, combined operations, reciprocals, rational vs irrational identification, and decimal approximation). Task A: compute the mean body temperature of patients at the Mathare North Health Centre dispensary using rational-number operations. Task B: calculate total and unit cost of githeri ingredients (maize and beans) for a school canteen serving 200 learners using combined operations on rational numbers. Task C: determine whether $\sqrt{5}$ m (a fence-post length on a Rift Valley farm) is rational or irrational, and approximate it to 4 significant figures. Groups present findings (3 minutes each), after which the class revisits the DQB, answering posted questions with accumulated evidence. The teacher cold-calls 6 learners to articulate connections back to the anchoring phenomenon.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** [Prime numbers] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Order of Real Numbers (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners individually read a brief stimulus card (displayed on screen / printed): 'At the Mathare North Health Centre dispensary, a nurse records three patient temperatures: 36.8 °C, 37.2 °C, and 37.6 °C. She also measures a fence post for the new dispensary waiting-room garden: the post must be exactly √5 metres long. Before calculating anything, predict: (a) Will the average temperature be a terminating decimal, a repeating decimal, or a never-ending non-repeating decimal? (b) Is √5 rational or irrational? Write your predictions and one-sentence justifications in your exercise books.' Learners write silently for 3 minutes, then share predictions with a seat partner (Think-Pair) for 1 minute.	Display the stimulus card and say: 'Read carefully and write your personal prediction — no discussion yet. You have 3 minutes.' [Set visible timer.] After 3 minutes: 'Now share your prediction with your partner. You have 60 seconds.' [Wait.] Cold-call 2 learners: 'Amina, what did you predict for the average temperature — and why?' [Allow 10-second WAIT TIME after the question before accepting an answer.] Then: 'Omondi, do you agree or disagree with Amina? Why?' Record both predictions (terminating / repeating / non-terminating non-repeating) on the board under the heading PREDICTIONS — we will check these at the end.	Think-Pair-Predict: activates prior knowledge from Lessons 1–6 and surfaces misconceptions before application tasks begin. Recording predictions publicly creates cognitive accountability — learners are motivated to verify whether their reasoning was correct.	Observable indicator: learner written predictions use the vocabulary 'terminating', 'repeating', or 'non-terminating non-repeating' and reference fraction equivalence or square-root irrationality. Teacher scans 4–5 books during the pair-share to gauge vocabulary use and conceptual accuracy before grouping for tasks.
Observe Phase 📹 **VIDEO:** Prime numbers Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Order of Real Numbers (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Groups of 4–5 receive a laminated task card with all three tasks and a calculator per group. They work simultaneously, rotating responsibility so each member leads one calculation and records results on a shared A3 group-answer sheet. TASK A — Average Body Temperature (Rational Numbers in Health): The Mathare North dispensary nurse records temperatures 36.8 °C, 37.2 °C, and 37.6 °C. (i) Express each as a fraction over 10. (ii) Compute the mean using combined operations: (36.8 + 37.2 + 37.6) ÷ 3. (iii) State whether the result is rational, and whether its decimal representation terminates or repeats. (iv) Connect your answer to the anchoring	Distribute task cards and calculators. Say: 'Each group has three tasks. Decide NOW who leads each task — the leader writes on the A3 sheet but everyone must be ready to explain any step. You have 15 minutes. Begin.' [Circulate continuously using a clipboard checklist of group names.] At the 5-minute mark, visit groups that appear stuck on Task C and prompt: 'Try squaring a few simple fractions — can you ever get exactly 5? What does that tell you?' [WAIT 10 seconds for a response before prompting further.] At the 10-minute mark, announce: 'Five minutes left — make sure	Structured Group Investigation with Role Accountability: assigning a named 'task leader' per task ensures distributed participation and prevents free-riding. The A3 recording sheet makes group thinking visible and provides an artefact for the presentation phase. Linking every task back to the Mathare North phenomenon maintains coherence across the lesson.	Observable indicators: (Task A) group correctly computes 111.6 ÷ 3 = 37.2 and states it is rational and terminating; (Task B) group obtains total cost = 200 × (0.15 × 45 + 0.08 × 120) = 200 × (6.75 + 9.60) = 200 × 16.35 = Ksh 3 270.00 and unit cost = Ksh 16.35; (Task C) group states √5 is irrational and writes √5 ≈ 2.236 (4 s.f.). Teacher flags any group writing √5 = 2.236 (equality sign) as a misconception to address in the Explain phase.

	phenomenon: why does this average 'end neatly'? TASK B — Githeri Canteen Costing (Combined Operations on Rationals): The canteen at Jomo Kenyatta High School in Kisumu serves githeri to 200 learners. Each serving requires 0.15 kg of dry maize (Ksh 45/kg) and 0.08 kg of dry beans (Ksh 120/kg). (i) Find the total mass of maize and beans needed. (ii) Compute the total ingredient cost for 200 servings. (iii) Find the cost per serving to the nearest 5 cents. (iv) Show that all intermediate values are rational numbers. TASK C — Fence Post Classification (Irrational Numbers in Construction): A farmer in the Rift Valley cuts a fence post to exactly √5 metres. (i) Is √5 rational or irrational? Justify by showing that no integer p and q exist such that √5 = p/q (or use the calculator evidence: display the decimal and inspect it). (ii) Use a calculator to find √5 correct to 4 significant figures. (iii) Explain in one sentence why the health worker's calculator showed a never-ending decimal for √2 when she measured the waiting-room tile diagonal. Groups have 15 minutes for all three tasks.	your A3 sheet has a sentence connecting each answer back to the nurse at Mathare North.' If a group finishes early, pose an extension: 'If the canteen raised the price of maize by 1/8, what is the new total cost? Is the answer still rational?'		
Explain Phase 📹 **VIDEO:** Prime numbers Source: Khan Academy (English - US curriculum) 🔍 Search ARES	Each group presents their A3 answer sheet to the class (3 minutes per group, rotating Task A → Task B → Task C across groups so each task is presented by a different group). After each	Manage presentations: 'Group 1, present Task A. You have 3 minutes — explain your steps, your answer, and your connection to the Mathare North nurse.' [Time strictly.] After Group 1: 'One	Gallery-style Group Presentations with Targeted Cold-Calling: public presentation raises the stakes of group accuracy and deepens individual accountability. Strategic cold-calling (6 named learners	Observable indicators: Cold-called learners use the phrases 'can be written as a fraction', 'terminating or repeating', and 'non-terminating non-repeating' correctly in their oral explanations. Teacher notes

for similar videos 📖 **HTML:** Order of Real Numbers (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	presentation, the class has 1 minute to ask one clarifying question or offer a correction. The teacher then facilitates a 5-minute whole-class synthesis discussion connecting all three tasks back to the anchoring phenomenon and to the sub-strand's key question 'Why are numbers important?'	question or correction from the class. Raise your hand.' Cold-call 1 learner. Repeat for Groups 2 (Task B) and 3 (Task C). During Task C presentation, if the group uses '=' instead of '≈' for √5, interject: 'Pause — I see you wrote √5 = 2.236. Is that exact? What symbol should we use and why?' [WAIT 10 seconds.] After all presentations, cold-call 3 more learners (total 6 cold-calls across the lesson) for synthesis: — Cold-call 4: 'Wanjiku, in Task A the average was 37.2 °C — a terminating decimal. In two sentences, explain WHY it terminates, using what you know about rational numbers.' — Cold-call 5: 'Kipchoge, can the canteen ever encounter an irrational number in their daily costing? Defend your answer.' — Cold-call 6: 'Nafula, connect the √5 fence post to the nurse's √2 tile diagonal. What do they have in common, and what does this tell us about the real-number system?' [Allow 10–12 seconds WAIT TIME after each cold-call question before accepting a response. Do not redirect immediately — allow productive struggle.] Close Explain phase with: 'So we now have strong evidence: numbers rooted in fractions — like prices, temperatures, masses — behave rationally and their decimals end or repeat. Numbers that arise from square roots of non-perfect squares — like √2 and √5 — are irrational and their	across the lesson) probes explanation and transfer rather than mere recall. The teacher's closing synthesis explicitly names the rational/irrational distinction as the resolution of the anchoring tension.	misconceptions (e.g., 'irrational means wrong') on a clipboard for targeted follow-up in Lesson 8.

		decimals go on forever without repeating. Both live on the same real-number line.'		
Driving Question Board (DQB) Creation 📹 **VIDEO:** Prime numbers Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Order of Real Numbers (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	The class Driving Question Board (first created in Lesson 1 and added to across Lessons 2–6) is displayed prominently. Each group receives three sticky notes (different colours: green = 'We can now answer this', yellow = 'We partially understand this', orange = 'Still puzzling us'). Learners spend 5 minutes reviewing the existing DQB questions and placing their coloured sticky notes against questions they can now address. One nominated group scribe then reads aloud each green-coded question, and the class collectively constructs a concise written answer (1–2 sentences) that the teacher writes on the board beside that question. Any new questions surfaced during today's tasks are added to the DQB on white sticky notes.	Uncover or display the DQB and say: 'Look at every question we have posted since Lesson 1. As a group, decide: which ones can you now answer with evidence from today or earlier lessons? Place a green sticky note on those. Yellow for partial, orange for still puzzling.' [Allow 5 minutes.] Then: 'Let's go through the green ones. Group 3 scribe — read the first green question aloud.' After the scribe reads, ask the class: 'Who can give a two-sentence evidence-based answer?' [Cold-call if no hand raised — this is cold-call 5 or 6 from the lesson total.] For any orange questions remaining, say: 'These are our unfinished business — we tackle them in Lesson 8. Write them in your exercise book as your personal investigation targets.' Add any new questions from today (e.g., 'Are there irrational numbers that are not square roots?') to the DQB, noting the date and lesson number beside them.	Colour-Coded DQB Consolidation: the three-colour system (green/yellow/orange) gives learners metacognitive agency — they self-assess their own understanding relative to the class's accumulated questions. Publicly constructing written answers to green questions transforms the DQB from a question repository into a growing evidence record, mirroring how scientists consolidate findings before drawing conclusions.	Observable indicators: the proportion of green vs orange sticky notes gives the teacher a rapid whole-class comprehension map. If fewer than 50% of questions receive green notes, the teacher adjusts Lesson 8 to revisit specific concepts. Teacher photographs the updated DQB for lesson planning reference.
Model Building Phase 📹 **VIDEO:** Prime numbers Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos	Learners individually update their personal Real-Number Model — a diagram they have been building since Lesson 1. The model must now include: (a) the two main branches of the real-number system (Rational and Irrational), each with at least two examples from today's tasks; (b) a label	Say: 'Open your exercise books to your Real-Number Model diagram. You have 4 minutes to update it with at least two examples from today — one from health or food (rational) and one from construction (irrational). Add your Big Idea sentence at the bottom.' [Display the Big Idea sentence	Cumulative Personal Model Revision with Peer Comparison: updating a model built across multiple lessons makes learning progression visible and tangible. The requirement to use a different-colour pen for new additions creates a visual record of conceptual growth over the	Observable indicators: the updated model contains labelled 'Rational' and 'Irrational' branches; examples from today's tasks appear in both branches; the Big Idea sentence explicitly mentions 'fraction', 'terminating or repeating', and 'non-terminating non-repeating'. Teacher spot-checks 5 models

📖 **HTML:** [Order of Real Numbers (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	showing where terminating decimals, repeating decimals, and non-terminating non-repeating decimals sit; (c) a one-sentence 'Big Idea' statement at the bottom answering the driving question: 'Some numbers from our everyday Kenyan lives end neatly because they can be written as exact fractions; others go on forever because they are irrational — and knowing the difference helps us choose the right tool (fraction, rounded decimal, or exact form) for better decisions.' Learners compare their updated model with a partner and annotate any new additions in a different colour pen.	starter on the board: 'Some numbers from our everyday Kenyan lives end neatly because…'] [After 4 minutes]: 'Compare your model with your partner. In one minute, identify one thing your partner added that you did not — and add it now in a different colour.' Close the lesson: 'Next lesson — Lesson 8 — we will use your models and the remaining DQB questions to write a full resolution to our driving question. Make sure your model is complete and legible. Your model is your evidence summary — treat it like a scientist's lab notebook.'	sequence. The Big Idea sentence forces synthesis at the level of the driving question rather than isolated facts.	during the partner-comparison minute and provides brief written feedback ('Strong — add the ≈ symbol for √5' or 'Good — can you add one more rational example from Task B?').

D. TEACHER REFLECTION
After the lesson, reflect on the following questions in your professional journal:\n1. COLD-CALL QUALITY: Did all 6 cold-called learners use precise vocabulary (rational, irrational, terminating, repeating, non-terminating non-repeating)? If not, which terms need reinforcement in Lesson 8?\n2. DQB COLOUR DISTRIBUTION: What percentage of posted questions received green sticky notes? If below 50%, which concepts (classification, combined operations, approximation) are least consolidated — and how will you adjust Lesson 8?\n3. TASK PERFORMANCE: Which of the three tasks (A, B, or C) produced the most errors or misconceptions? Note specific errors (e.g., using '=' instead of '≈' for √5, or incorrect order of operations in Task B) to address explicitly in Lesson 8.\n4. PHENOMENON CONNECTION: Did learners independently connect their task results back to the Mathare North nurse without teacher prompting? If not, consider opening Lesson 8 with a direct re-reading of the phenomenon to sharpen the link.\n5. EQUITY CHECK: Were there learners who did not contribute during group tasks or presentations? Plan deliberate roles for them in Lesson 8's final resolution activity.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners computed the mean body temperature of three Mathare North Health Centre patients (Tasks A), calculated total and unit githeri ingredient costs for a 200-learner school canteen in Kisumu (Task B), and determined that √5 m (a Rift Valley farm fence-post length) is irrational, approximating it to 4 significant figures (Task C).
What did I learn?	Learners confirmed that numbers arising from everyday Kenyan contexts — health temperatures, food costs, masses — are rational because they can be expressed as exact fractions, giving terminating or repeating decimals; whereas numbers arising from square roots of non-perfect squares (like √2 and √5) are irrational, producing non-terminating non-repeating decimals. The real-number system contains both types.

How does this explain the phenomenon?	Learners explained, through group presentations and targeted cold-calling, why the Mathare North nurse's average temperature 'ended neatly' (rational arithmetic on terminating decimals always yields a rational result) while the tile-diagonal calculation displayed a never-ending decimal ($\sqrt{2}$ is irrational). They updated their DQB and personal Real-Number Models to consolidate evidence from all seven lessons.

LESSON 8 (40 minutes): Model Building, Synthesis & Final Explanation — Completing the Real Numbers Reference Map

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners individually construct a finalised 'Real Numbers Reference Map' that synthesises all sub-strand learning, answer the driving question in their own words, and complete a short written test as portfolio evidence.
Knowledge	– The real-number system forms a nested hierarchy: $\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R}$, with irrational numbers ($\mathbb{R} \setminus \mathbb{Q}$) sitting alongside the rationals inside □. – A rational number can always be expressed as p/q ($p, q \in$ □, $q \neq 0$) giving a terminating or repeating decimal; an irrational number cannot, giving a non-terminating, non-repeating decimal. – Combined operations on rational numbers always yield a rational result (closure); the reciprocal of any non-zero real number x is $1/x$, and $x \times (1/x) = 1$.
Skills	– Construct and annotate a hierarchical model of the real-number system populated with Kenyan-context examples, including at least one combined-operations worked example and one reciprocal example. – Evaluate and revise an initial prediction (Lesson 1 model) against accumulated evidence to produce a coherent final explanation for the driving question.
Attitudes	– Appreciate that mathematics provides precise language for describing everyday Kenyan numerical experiences — from health-centre readings at Mathare North to tile measurements — and that this precision supports better real-life decisions. – Show responsibility and integrity by independently completing the written test and portfolio artefact without copying.
Key Inquiry Question	Why do some numbers from our everyday Kenyan lives 'end neatly' while others go on forever — and how does knowing the difference help us make better decisions?
Purpose in Storyline	This is the culminating lesson in which all threads of the investigation — classification, combined operations, reciprocals, and real-life application — are woven together. Learners revisit Sister Wanjiku's phenomenon from Lesson 1, compare their naive Lesson-1 explanation with their current understanding, finalise their cumulative model as a portfolio artefact, and demonstrate mastery through a short written test.
Safety Notes	No specific hazards. Remind learners that calculators are shared resources; handle with care and return them to the charging tray after use (Value: Responsibility).

B. LESSON OVERVIEW

Lesson 8 is the capstone of the Real Numbers sub-strand. It opens by returning learners directly to the anchoring phenomenon: Sister Wanjiku at Mathare North Health Centre dividing three temperatures $(36.8 + 37.0 + 37.5) \div 3$ to get a neat terminating decimal, yet finding that the diagonal of the waiting-room tile produces $\sqrt{2} \approx 1.41421356\ldots$, a decimal that never ends and never repeats. Having spent seven lessons gathering evidence — classifying numbers, performing combined operations, exploring reciprocals, and applying real numbers in Kenyan contexts — learners are now equipped to explain this phenomenon fully and precisely.

In the first half of the lesson learners individually construct their "Real Numbers Reference Map": a hand-drawn, hierarchically annotated diagram showing $\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset$ □ with irrationals sitting alongside the rationals. Each map must include at least three Kenyan-context examples per set, one fully worked combined-operations example (using BODMAS with fractions drawn from market or health data), and one reciprocal example with verification. This model-building activity serves simultaneously as a sensemaking tool and a portfolio artefact.

The second half of the lesson involves a structured DQB closure discussion — comparing the class's Lesson-1 sticky notes with their current answers — followed by a short five-question written test (the formal portfolio assessment mandated by KICD). The lesson closes with learners writing a two-sentence "final answer" to the driving question in their own words, grounding it explicitly in Sister Wanjiku's data. Teacher reflection questions are provided to guide post-lesson professional practice.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** Prime numbers Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Order of Real Numbers (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners open their exercise books to their Lesson-1 entry task — the rough sketch or sentence they wrote in response to: 'Why does 36.8 + 37.0 + 37.5 ÷ 3 end neatly, but √2 does not?' They read their own Lesson-1 answer silently for 90 seconds, then write ONE sentence below it beginning: 'Now I think the real reason is…' This activates prior knowledge and creates a visible before/after comparison without yet revealing the final answer to peers.	Project or write on the board the original phenomenon text: 'Sister Wanjiku enters 36.8 + 37.0 + 37.5 ÷ 3 and sees a tidy decimal. She measures the tile diagonal and sees 1.41421356… Why?' Say exactly: 'Take 90 seconds — silent, individual — to re-read what you wrote on Day 1. Then write one new sentence starting with the words: NOW I THINK THE REAL REASON IS…' Set a visible timer for 90 seconds. After time, say: 'Keep your books open — we will come back to compare at the end of the lesson.' Do NOT ask for sharing yet. Cold-call 0 learners at this stage — the predict is entirely private to preserve honest self-assessment.	Elicit–Compare–Revise (ECR): By surfacing the Lesson-1 naive explanation before instruction, learners create a cognitive anchor. The contrast between their old and new sentences at the end of the lesson makes conceptual growth visible and personally meaningful, reinforcing metacognitive awareness.	Teacher circulates during the 90 seconds and scans (without reading aloud) whether Lesson-1 sentences mention any mathematical language (fraction, decimal, square root). This baseline check informs how much scaffolding to provide during the Map-building phase. Look for: Are learners referencing p/q, terminating, irrational — or are sentences still vague ('some numbers are weird')? Adjust explanatory depth in the Explain phase accordingly.
Observe Phase 📹 **VIDEO:** Prime numbers Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Order of Real Numbers (Flexbook) Source: CK-12 🔍 Search ARES	Learners receive a single A4 'Reference Map Starter Sheet' (or draw the template in their books). The sheet shows five empty nested ovals labelled from innermost to outermost: Natural Numbers (ℕ), Integers (ℤ), Rational Numbers (ℚ), Irrational Numbers (ℝ \ ℚ), Real Numbers (□). Each oval has three blank bullet lines for examples. Learners spend 8 minutes working individually and silently to populate every oval with at least three Kenyan-context examples drawn	Distribute or display the Reference Map template. Say: 'You have 8 minutes — working alone — to fill every oval with at least three examples from our lessons. Every example must come from a Kenyan context we used. For the worked examples on the right side: Example A — use the temperature data: evaluate (36.8 + 37.0 + 37.5) ÷ 3, show every BODMAS step, and state whether the answer is rational or irrational and why. Example B — the blood pressure ratio 120/80 simplifies to 3/2; find	Evidence-to-Model Mapping: Learners are not passively copying definitions; they are selecting and placing evidence from seven lessons of Kenyan-context data into a formal mathematical structure. This strategy (adapted from NGSS Science and Engineering Practice 2 — Developing and Using Models) makes the abstract number hierarchy concrete by anchoring each set to lived experience (shoe sizes, body temperatures, tile diagonals, maize prices).	Teacher looks for three specific indicators during circulation: (1) Correct nesting — natural numbers inside integers inside rationals, irrationals NOT inside rationals but both inside reals. Flag any learner who places √2 inside ℚ — this is the key misconception from Lesson 2. (2) Kenyan examples — generic 'any fraction' is insufficient; teacher prompts: 'Give me a specific number from our lessons.' (3) BODMAS steps in Example A — the common error is dividing only 37.5 by 3 rather than

for similar readings	from any of Lessons 1–7. They may consult their exercise-book notes. Alongside the map they copy and complete two structured worked examples: (A) a combined-operations problem using Sister Wanjiku's temperature data, and (B) a reciprocal verification using the blood-pressure reading 120/80 simplified.	its reciprocal, verify by multiplication, and state what set it belongs to.' Circulate continuously. After 4 minutes give a mid-point prompt: 'Check — does your □ oval include negative integers? Does your Irrational oval include at least one square root and π?' At the 8-minute mark say: 'Pens down on the map for now — we will add one more annotation in the Explain phase.'		the full sum; check that learners write (36.8 + 37.0 + 37.5) ÷ 3 = 111.3 ÷ 3 = 37.1 with the bracket step shown explicitly.
Explain Phase 📹 **VIDEO:** Prime numbers Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Order of Real Numbers (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners add a final annotation to their Reference Map: a two-colour 'Decision Arrow' labelled with the two questions they now know to ask about any number: (1) 'Can it be written as p/q with $p, q \in \mathbb{Z}$ and $q \neq 0$?' → YES = Rational; (2) 'Does its decimal terminate or repeat?' → YES = Rational. They then write their Final Explanation sentence directly on the map: 'Sister Wanjiku's temperature average ends neatly because it is RATIONAL — it equals the fraction 371/10, which terminates. The tile diagonal $\sqrt{2}$ goes on forever because it is IRRATIONAL — no fraction p/q ever equals it exactly.' Three volunteers share their phrasing aloud; class refines wording collaboratively.	Say: 'Let us now add the explanation layer to your map. Draw two arrows from the outer □ oval — one pointing to $\mathbb{Q}$, one pointing to the irrational region. Label each arrow with the test question we established in Lesson 3.' Give WAIT TIME of 60 seconds for learners to draw. Then say: 'Now, on the bottom of your map, write — in your own words — a two-sentence answer to our driving question. Your sentence MUST name Sister Wanjiku, MUST include the word rational or irrational, and MUST reference the fraction p/q definition.' Allow 2 minutes of silent writing. After 2 minutes, cold-call 3 learners by name (choose one high-confidence, one mid-confidence, one who struggled in Lessons 2–3 based on earlier formative data). For each, say: 'Read your sentence exactly as written.' After each reading, ask the class: 'Thumbs up if their sentence correctly used p/q — thumbs sideways if something is missing.' Correct any sentence that says 'irrational numbers are bigger' or 'they are harder' — redirect to: 'The difference is about FORM, not size or difficulty. Can it be written as a fraction of two integers? That	Claim–Evidence–Reasoning (CER) Framework: Learners are required to state a Claim (the number is rational/irrational), cite Evidence (the fraction form or the non-terminating decimal), and provide Reasoning (linking to the p/q definition from Lesson 3). This three-part structure, made explicit on the Reference Map, scaffolds scientific explanation writing and directly addresses the KICD core competency of Critical Thinking and Problem Solving.	Listen for three quality indicators in the three cold-called sentences: (1) The word 'rational' or 'irrational' is used correctly — not 'normal' or 'weird'. (2) The p/q definition is referenced, not just 'it terminates'. (3) The sentence is specifically about Sister Wanjiku's data, not a generic statement. If fewer than 2 of the 3 cold-called learners meet all three indicators, pause and do a whole-class choral response: 'Everyone together — a rational number is any number that can be written as…?' (Expected response: 'p over q, where p and q are integers and q is not zero.')

		is the only question that matters.'		
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Prime numbers] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Order of Real Numbers (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	The teacher reveals the class Driving Question Board from Lesson 1 (physical sticky notes or projected digital board). Learners compare the unanswered questions from Lesson 1 with what they now know. Working in pairs for 3 minutes, learners identify: (a) one question from the DQB that is now fully answered, (b) one question that is partially answered, and (c) one new question that has emerged from the sub-strand (e.g., 'Are there more irrationals than rationals?' or 'Why does π never repeat?'). Each pair reports one fully-answered question to the class. Teacher physically moves or stamps the answered sticky notes as 'RESOLVED'. The board now shows the complete intellectual journey from Lesson 1 to Lesson 8.	Uncover or project the Lesson-1 DQB. Say: 'This board has been with us since Day 1. Eight lessons ago, we stood here with questions and a mystery. Let us see what we have resolved.' Point to two or three specific Lesson-1 sticky notes (e.g., 'Why does the calculator show so many digits for √2?', 'Is 22/7 the same as π?'). Ask: 'With your partner — 3 minutes — decide: fully answered, partly answered, or still open? And add one NEW question our learning has sparked.' After 3 minutes, cold-call 4 pairs (choose different pairs from the Explain phase). For each 'fully answered' question, ask: 'In one sentence — what is the answer?' After all pairs report, say: 'Look at your Lesson-1 sticky note one more time. How different is your thinking now from Day 1?' Allow WAIT TIME of 15 seconds. Do not cold-call — this is a private moment of metacognitive reflection. Then say: 'The new questions you are asking — like whether there are more irrationals than rationals — show that you have not just learned facts. You have become mathematical thinkers. Those questions will be answered in future mathematics courses.'	Driving Question Board Closure (DQB-C): By physically resolving sticky notes and comparing Lesson-1 questions against current understanding, learners experience the full arc of scientific inquiry — from puzzlement to explanation. This strategy makes the growth in understanding tangible and communal, reinforcing that mathematics is a process of building knowledge over time, not a collection of disconnected facts.	Teacher checks whether the 'fully answered' questions offered by pairs are genuinely resolved with mathematical precision (e.g., 'The calculator shows many digits for √2 because √2 is irrational — its decimal never ends or repeats, so the calculator can only show an approximation') versus vague responses ('because it's complicated'). If pairs offer vague answers, prompt: 'Can you use the word irrational and the definition p/q in your answer?' Also note which learners generate high-quality 'new questions' — these are candidates for extension tasks or mathematics club activities.
Model Building Phase 📹 **VIDEO:** [Prime numbers] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos	Learners spend the final 10 minutes in two parts. Part 1 (6 minutes): Individual silent written test — 5 focused questions printed on a slip or written on the board, covering: (Q1) classifying three numbers as rational or irrational with justification; (Q2) a combined-operations problem with BODMAS steps shown; (Q3) finding and	Say: 'Your Reference Map is now your portfolio artefact for this sub-strand — it belongs to you and shows your mathematical thinking across eight lessons. Before you submit it, sign and date it. You are claiming this work.' Display the 5 test questions. Say: 'You have 6 minutes — this is individual and silent. No calculators for Q1, Q2,	Cumulative Model Revision (CMR): The Reference Map is not a summary worksheet — it is a living model that has been built layer by layer across the unit, mirroring the NGSS Science and Engineering Practice of Developing and Using Models. In Lesson 8 the model receives its final annotations (Decision Arrows,	Collect and review the 5-question test slips before the next lesson. Use the following observable indicators: Q1 — learner correctly identifies rational vs irrational AND provides the p/q justification (not just 'it terminates'); Q2 — BODMAS order is explicit, fractions converted before operations, final answer stated as

📖 **HTML:** [Order of Real Numbers (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	verifying a reciprocal; (Q4) placing four numbers in the correct set on a mini nested-oval diagram; (Q5) one-sentence answer to the driving question using the p/q definition. Part 2 (4 minutes): Learners sign and date their Reference Map, write their name clearly, and place it in the class portfolio tray. They also return to their Lesson-1 predict sentence in their exercise book and circle or highlight the biggest change in their thinking.	Q3 — mental and written arithmetic only. You may use a calculator only for Q4 verification.' Read all five questions aloud once slowly, then say: 'Begin.' Circulate without commenting on answers — only redirect off-task behaviour. At the 6-minute mark say: 'Pens down.' Collect test slips. Then say: 'Final 4 minutes: sign your Reference Map, date it, and write at the very top: MY ANSWER TO THE DRIVING QUESTION: — and complete that sentence in your best mathematical language. Then place the map in the portfolio tray.' As the bell approaches, close with: 'Sister Wanjiku's numbers puzzled us on Day 1. Today you can explain exactly why some numbers end neatly and others do not — and more importantly, you can use that knowledge to make better decisions about the numbers in your own Kenyan lives. Well done.'	Final Explanation sentence, driving-question answer), transforming it from a classification chart into a genuine explanatory tool. The side-by-side visibility of Lesson-1 naive sentences versus Lesson-8 precise explanations makes conceptual change concrete and assessable.	rational/irrational; Q3 — reciprocal stated, product shown = 1, set membership stated; Q4 — nested diagram correct with irrationals outside ℚ but inside ℝ; Q5 — driving-question sentence includes 'rational', 'p/q', and a Kenyan example. Score each question as Meets / Approaches / Below Expectations using the KICD rubric language. Flag any learner who places √2 in ℚ on Q4 for targeted follow-up in the next sub-strand (Indices).

D. TEACHER REFLECTION

1. When learners compared their Lesson-1 naive sentence with their Lesson-8 explanation, what specific language shifts did you observe — did they move from everyday descriptions ('the number is messy') to precise mathematical language ('non-terminating, non-repeating decimal that cannot be expressed as p/q')? Which learners still used informal language and what support will you provide in Sub-Strand 1.2?

2. On the 5-question written test, which question produced the most errors? If Q2 (combined operations BODMAS) was the most common stumbling block, consider opening Sub-Strand 1.2 (Indices) with a brief BODMAS warm-up that bridges from rational number operations to index notation.

3. Did the Reference Map models submitted to the portfolio show correct nesting ($\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R}$ with irrationals correctly positioned outside $\mathbb{Q}$)? If more than 25% of maps showed incorrect placement of irrational numbers, this misconception will need to be revisited when index numbers and surds appear in later strands.

4. During the DQB closure, how sophisticated were the 'new questions' that learners generated? Questions like 'Are there more irrational numbers than rational numbers?' or 'Can you add a rational and irrational and get a rational?' indicate deep engagement with mathematical structure and should be celebrated publicly and flagged for the Mathematics Club or extension tasks.

5. Reflect on the equity of participation across the eight lessons: which learners contributed most to whole-class discussion, and which were consistently silent? Does the portfolio evidence (Reference Map + test) reveal hidden competence in quieter learners that was not visible in oral phases? How will you use portfolio data to ensure every learner's growth is recognised and reported accurately?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners observed their own Lesson-1 naive explanations alongside their Lesson-8 precise mathematical explanations, making their conceptual growth visible. They observed that the Reference Map — populated entirely with Kenyan-context numbers from health, agriculture, and everyday life — forms a coherent nested hierarchy ($\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R}$, with irrationals alongside $\mathbb{Q}$ inside $\mathbb{R}$). They also observed, through the DQB closure, that every question the class asked on Day 1 about Sister Wanjiku's calculator display now has a precise mathematical answer.
What did I learn?	Learners learned that the real-number system is a complete, nested structure where every number encountered in everyday Kenyan life — body temperatures, shoe sizes, blood pressure ratios, market prices, tile diagonals — can be precisely located. Rational numbers (expressible as p/q, $q \neq 0$) produce terminating or repeating decimals and are closed under all four arithmetic operations. Irrational numbers ($\sqrt{2}$, π, $\sqrt{5}$) cannot be expressed as p/q and produce non-terminating, non-repeating decimals. The reciprocal of any non-zero real number x is $1/x$ with $x \times (1/x) = 1$. Knowing which type of number you are working with determines whether a calculation 'ends neatly' or requires an approximation — a distinction with direct consequences for decision-making in health, construction, and commerce.
How does this explain the phenomenon?	The phenomenon is now fully explained: Sister Wanjiku's temperature average $(36.8 + 37.0 + 37.5) \div 3 = 111.3 \div 3 = 37.1$ ends neatly because all three temperatures are rational numbers (e.g., $36.8 = 368/10$), and rational numbers are closed under addition and division by a non-zero integer — the result $37.1 = 371/10$ is itself rational, with a terminating decimal. The tile diagonal $\sqrt{2}$ goes on forever because 2 is not a perfect square: no fraction p/q (with $p, q \in \mathbb{Z}$, $q \neq 0$) satisfies $(p/q)^2 = 2$ exactly, so $\sqrt{2}$ is irrational and its decimal is non-terminating and non-repeating. Knowing this difference helps Sister Wanjiku — and any Kenyan decision-maker — understand when a calculator result is exact and when it is merely an approximation that must be treated with appropriate caution.

DIFFERENTIATION AND INCLUSION		
Learner Need	**Support Strategies**	**Extension Strategies**
English Language Learners	Provide bilingual glossaries; allow use of first language for initial model drawing.	Write extended explanations of observations; lead group discussions in English.
Students with learning difficulties	Provide structured note frames; pair with a supportive partner during investigations.	Mentor peers; create additional visual models to explain concepts.
Gifted and advanced learners	Access to additional reading materials; challenge questions during DQB.	Lead model-building discussions; research and present extension topics to class.